Highlights

Causality-Inspired Models for Financial Time Series Forecasting

- We systematically compare causal versus predictive feature selection for financial forecasting.
- Invariance-based methods identify predictors that remain stable across market regimes.
- Invariant-based methods demonstrate superior robustness and lower error during periods of market crisis.
- Trading strategies based on invariant features yield higher risk-adjusted returns under stress.

Causality-Inspired Models for Financial Time Series Forecasting

Abstract

Causality is a central concept in econometrics, yet its role in economic and financial time series forecasting remains contested. While traditional models, both econometric and machine learning-based, often rely on non-causal or spurious relationships to achieve high predictive accuracy, recent research has begun to question the robustness of such approaches. Motivated by this tension, we examine the practical value of causality in forecasting asset returns. Adopting an operational perspective grounded in the concept of invariance, we compare feature selection methods that leverage causal reasoning with those based purely on predictive performance. Through a comprehensive empirical study, we find that causality-inspired methods yield more robust forecasts, particularly in volatile market conditions. To the best of our knowledge, this is the first study to systematically evaluate the forecasting benefits of causal techniques in a financial time series context.

Keywords:
Causal Discovery, Invariant Prediction, Robust Forecasting, Financial Time Series, Regime Shifts, Feature Selection

1. Introduction

Financial time series forecasting plays a vital role in investment strategies, risk management, and economic planning. However, it remains a challenging task due to the complexity, noise, and inherent volatility of financial markets. Traditional forecasting models often fail under distributional shifts, leading to suboptimal decisions in dynamic environments [1, 2, 3, 4]. These models typically focus on minimizing prediction error over historical data, which may not generalize when underlying distributions change. This challenge is particularly pronounced in financial time series involving a large cross-section of assets with relatively few observations at low to mid frequencies (e.g., quarterly, monthly, or daily data). [5] shows that complex models often fail to outperform simpler, more robust alternatives. (EC1) This highlights a central challenge in financial forecasting: designing models that remain reliable under distributional instability.

From both the econometric and machine learning perspectives, the traditional forecasting literature has primarily concentrated on the prediction step, often through the development of increasingly complex models. In machine learning, feature selection is typically handled implicitly by the model architecture, such as tree-based models or neural networks, with limited emphasis on domain-driven variable curation [6, 7, 8, 9].In contrast, econometric approaches have relied on a combination of explicitly curated variable sets, guided by economic theory and financial intuition [10, 11], and regularization-based models, such as LASSO, which perform implicit variable selection [12, 13]. (EC1) However, neither perspective has placed sufficient emphasis on identifying *features that remain structurally stable under varying market conditions.*

Causality-inspired models offer a promising alternative by focusing on features that remain stable in varying market conditions. To this end, [14] highlights the role of causal reasoning as one alternative to achieving robustness under distributional changes, motivating a new class of forecasting approaches that balance predictive performance with regime stability. Although causal methods have gained traction in cross-sectional tasks [15, 16, 17], their application to time series forecasting, particularly in finance, remains limited [18].

(EC1) It is important to distinguish the notion of *stability* considered in this paper from the concept of stability traditionally studied in the forecasting literature. In that context, stability is generally understood as the consistency of forecasts under small perturbations of the information set, such as changes in the estimation sample, forecast origin, or model specification. This includes notions such as the smooth evolution of forecasts over time, limited sensitivity to new observations, and robustness to re-estimation procedures, all of which aim to reduce excessive variability in model outputs and improve decision-making reliability [19].

In contrast, the notion of stability we consider is structural: it concerns the invariance of the predictive relationship between features and the target across different market regimes. In this sense, our notion of stability is closely aligned with the concept of robustness as defined in the complex systems and biological literature, namely the ability of a system to maintain its function despite changes in its internal structure or external environment [20, 21].

Rather than focusing on the stability of forecasts conditional on a model, our objective is to identify features whose predictive power itself remains stable under distributional shifts. This distinction is particularly important in financial applications, where robust performance depends not only on smooth forecasts, but on the persistence of the underlying economic relationships driving those forecasts.

This perspective motivates the central question of this study: whether

it is possible to move beyond stability at the level of forecasts and instead achieve stability at the level of the underlying predictive structure. In this spirit, we investigate the use of causal reasoning in time series forecasting, with a particular focus on financial applications. Specifically, we explore invariant prediction techniques as a means of identifying robust features to forecast asset returns. These methods aim to improve robustness in the face of concept drift and distributional change by selecting features that are stable across environments [22].

Main contributions. This paper contributes to the literature along two closely related dimensions. First, we extend the application of causality-inspired prediction methods from cross-sectional settings to financial time series forecasting. Specifically, we provide a systematic empirical comparison between state-of-the-art causal discovery algorithms and traditional, non-causal feature selection techniques for return prediction, offering new evidence on the relative merits of each approach in a domain where causal methods remain largely unexplored.

Second, and more importantly, we address the problem of forecast robustness under distributional instability. Building on the notion of robustness as the ability of a system to maintain its function under changing environments, we show that causality-inspired feature selection provides a principled way to construct predictive models that remain reliable when the underlying data-generating process evolves over time. By evaluating predictive performance across distinct market regimes, we demonstrate that features selected through invariant prediction techniques yield forecasts that are not only competitive in-sample but also more stable out-of-sample relative to those obtained from conventional approaches.

(RC1) To operationalize this comparison, we deliberately adopt a simple linear regression framework estimated using ordinary least squares (OLS). This design choice is intended to isolate and evaluate the added value of causality-inspired feature selection compared to standard predictive approaches, using forecast performance as the primary criterion. By holding the prediction model constant across all methods, we ensure that any observed differences in performance can be attributed to the feature selection step rather than to model complexity.

Following [23], we adopt the assumption that invariant predictors are closely related to causal variables. Causal discovery algorithms aim to estimate the direct causal parents of a target variable under varying assumptions. In the literature on graphical models, it is well established that the conditional distribution of a variable, given its direct causes, remains invariant to interventions in other variables [23]. Motivated by this, we define our **invariance-based feature selection methods** (IBFS) as those that use causal discovery algorithms to select features.

In contrast, we benchmark against **prediction-based feature selec-**

tion methods (PBFS), which prioritize variables solely based on their contribution to minimize prediction error. Sequential feature selection serves as a representative baseline: while relatively simple, it offers a clear contrast by optimizing a criterion that is misaligned with invariance.

To isolate the predictive value of the selected features, we use a common linear regression model estimated using ordinary least squares (OLS) across all methods. This allows for a fair comparison between IBFS and PBFS, independent of model complexity.

We evaluated these methods using two datasets. The first, following [18], consists of monthly macroeconomic and financial time series for the Eurozone and Switzerland, where the task is to forecast exchange rate returns of EUR/CHF. The second involves predicting monthly returns of the SPDR S&P 500 ETF (SPY) using macroeconomic predictors curated by [24] from the Federal Reserve Bank of St. Louis.

(EC1) We hypothesize that the primary benefit of causality-inspired feature selection lies in improving forecast robustness under distributional shifts, particularly during periods of structural change in the underlying data-generating process. Previous work by [22] suggests that causality-inspired models can enhance the robustness of predictive systems, especially under structural changes or nonstationarities.

To empirically assess this potential advantage, it is essential to define, in a principled way, the periods during which dataset shifts occur. Fortunately, financial time series naturally experience such shifts due to exogenous shocks, such as the COVID-19 pandemic or the Global Financial Crisis (GFC), as well as through more frequent volatility spikes. These episodes provide a practical framework for labeling periods of instability, enabling a clearer evaluation of whether causality-based feature selection offers superior robustness compared to purely predictive alternatives.

Our results show that IBFS methods achieve predictive accuracy comparable to purely predictive approaches, but with fewer predictors. Importantly, causality-inspired methods exhibit significantly lower relative prediction errors during periods of increased volatility, such as the GFC, the COVID-19 outbreak, and the EUR/CHF peg break, highlighting their robustness under dataset changes. Furthermore, we empirically demonstrate the economic relevance of these findings by implementing trading strategies for SPY and EUR/CHF based on the predicted return directions. The resulting strategies yield higher relative risk-adjusted returns, as measured by Sharpe and Sortino ratios, compared to those based on PBFS methods.

Paper structure. The remainder of this paper is organized as follows:. Section 2 reviews our contributions to the finance literature and causal discovery. Section 3 introduces the proposed forecasting framework. Section 4 describes the data source. Section 5 explains our setup of empirical experiments. Section 6 presents the empirical results of return forecasting.

Finally, Section 7 summarizes the results and discusses limitations and future research directions.

2. Related Work

Forecasting asset returns is a continuously evolving area of research with a long-standing history, attracting the interest of both practitioners and academics. The statistical properties of return time series—such as low signal-to-noise ratios, heteroskedasticity, and non-stationarity—render forecasting tasks particularly challenging [25, 26].

The traditional econometric literature has proposed a variety of linear models, including autoregressive integrated moving average (ARIMA) [27, 28] and vector autoregression (VAR) [29, 30], which incorporate lagged returns and external regressors for prediction. More recently, there has been a growing body of literature exploring nonlinear machine learning approaches for return forecasting [31, 32, 33]. For instance, [34] conducts a comprehensive analysis comparing the predictive performance of models ranging from simple linear regressions to neural networks, using 94 firm characteristics as input features for monthly equity return forecasting.

A substantial portion of the literature has identified individual predictors with statistically significant forecasting power based on rigorous in-sample testing. These include, but are not limited to, valuation ratios [35, 36, 37, 38, 39, 40], historical returns and technical indicators [41, 42, 43, 44, 45, 46], macroeconomic variables [47, 48, 49], and market microstructure indicators [50, 51, 52, 53, 54]. [55] provides a thorough evaluation of the predictive power of published predictors, emphasizing the importance of out-of-sample testing and advocating for the continued evaluation of new variables. Subsequent studies [56, 45, 46] reinforce the value of these predictors and introduce novel ones.

Despite the abundance of potential predictors, appropriate feature selection remains critical for robust forecasting [57, 58]. Effective feature selection can mitigate overfitting, enhance out-of-sample performance, and improve interpretability. Common approaches include regularization-based models such as LASSO [59, 60], correlation-based selection methods [61], and other algorithmic selection techniques.

The machine learning literature on causal inference has traditionally drawn a clear distinction between prediction and causality. As emphasized by Pearl [62] in his ladder of causation, predictive models operate at the level of associations, while causal models concern themselves with interventions and counterfactual reasoning. From this perspective, the goals of predictive modeling and causal inference appear to be in conflict: the former seeks statistical correlations to minimize prediction error, whereas the latter aims to uncover structural mechanisms that remain stable across varying environments.

The machine learning literature on causal inference has traditionally drawn a clear distinction between prediction and causality. As emphasized by [62] in his ladder of causation, predictive models operate at the level of associations, while causal models concern themselves with interventions and counterfactual reasoning. (EC1) As discussed in Section 1, the notion of stability adopted in this paper is structural, referring to the invariance of the predictive relationship across environments, rather than the smoothness of forecast outputs.

A practical implication of this divide, particularly in the context of feature selection, is the assumption that predictive models only require the Markov blanket of the target variable, without needing to distinguish direct causes (parents) from child and spouse nodes. However, this view began to shift with the introduction of invariant causal prediction (ICP) by Peters et al. [23], who proposed leveraging the invariance property of causal relationships for feature selection. They demonstrated that the set of direct causes of a target variable remains invariant under a variety of interventions, and this invariance can be exploited to make a causal discovery. The ICP framework thus offered one of the first practical ways to integrate causal principles into predictive modeling through feature selection. This idea was later extended to time series data by Pfister et al. [18], who applied analogous principles of invariance to sequential settings.

The connection between causal feature selection and predictive performance was more explicitly developed by Pfister et al. [63], who argued that identifying the causal parents of a target variable, although potentially detrimental to in-sample performance, can lead to superior generalization in the presence of distributional changes. Their results suggest that causality-inspired feature selection can offer significant robustness advantages, especially when the test environment differs from the training environment.

The key insight behind this robustness lies in the invariance property of causal parents [23]: when selecting predictors based on their stability across different environments or interventions, the set of direct causes, denoted $\mathrm{PA}(Y)$, remains statistically invariant. This means that the conditional distribution of the target variable Y, given its causal parents, remains unchanged even when other variables in the system are perturbed. In contrast, non-causal variables (such as descendants of Y or variables connected via non-causal paths) may induce spurious correlations that vary across environments. These spurious or unstable associations, often introduced through indirect paths that are sensitive to intervention, can degrade generalization when carried into predictive models. The ICP framework capitalizes on this by identifying submodels that are invariant across environments; with high probability, the causal parent set will be among these. Moreover, this principle is not limited to ICP alone: any causal discovery

method that aims to recover the underlying causal graph, under suitable assumptions, can be used as a feature selector in predictive tasks. These approaches offer a principled, practical mechanism for incorporating causal reasoning into model construction, thereby improving performance in scenarios where training and test distributions may differ.

3. Forecasting Framework

Our forecasting framework consists of two main steps: feature selection and forecasting based on the selected features. The overarching goal is to assess the robustness and accuracy of forecasting models that rely on either invariance-based or prediction-based feature selection methods. More specifically, we aim to quantify the marginal contribution of causality-inspired ideas, proxied by invariance-based methods, to forecasting performance.

This focus implies that the richness of our experimental design lies primarily in the diversity of IBFS (invariance-based feature selection) models, rather than in the complexity of the final-step forecasting models. In other words, we deliberately use simple forecasting models in the final step in order to isolate and evaluate the impact of the feature selection process itself.

We begin Section 3.1 by formally defining the forecasting problem and introducing the necessary notation. In Section 3.2, we define the class of IBFS methods and describe the causal discovery methods we use as proxies for this class. In Section 3.4, we formalize the class of PBFS (prediction-based feature selection) methods.

The feature selection literature is vast, yet it rarely classifies methods along the invariance–prediction spectrum. To support our proposed classification, we introduce a set of desirable properties that PBFS methods should satisfy in order to serve as an effective counterpoint to IBFS methods in answering our core research questions. Once these properties are established, we describe in detail the specific PBFS methods used in our experiments that align with these criteria.

3.1. Notation

Let $\{(Y_t, \boldsymbol{Z}_t')'\}_{t=1}^{\infty}$ be a multivariate time series process taking values in $\mathbb{R}^{d+1}$, where $Y_t \in \mathbb{R}$ is a scalar target (response) variable and $\boldsymbol{Z}_t \in \mathbb{R}^d$ is a d-dimensional vector of observed covariates at time t. The forecasting task consists of predicting Y_{t+h} for horizon(s) $h = 1, \ldots, H$.

For integers $p \geq 1$ and $r \geq 0$, we define the n-dimensional feature vector

$$\boldsymbol{X}_t := (Y_{t-1}, \ldots, Y_{t-p}, \boldsymbol{Z}_t', \ldots, \boldsymbol{Z}_{t-r}')',$$

where $n = p + d(r + 1)$. This vector collects both lags of the target and lags of the covariates, and serves as the input to our forecasting models.

Throughout the remainder of the paper, we fix the forecasting horizon to $h = 1$ and focus exclusively on one-step-ahead prediction tasks. We denote the prediction or estimate of Y_{t+1} at time t by $\hat{Y}_{t+1|t}$. When evaluating the marginal contribution of a specific feature $X_t^j \in \boldsymbol{X}_t$, we denote by $\boldsymbol{X}_t^{-j}$ the feature vector with the j-th component removed, and by $\hat{Y}_{t+1|t}^{-j}$ the corresponding forecast obtained when excluding this feature.

3.2. Invariance-Based Feature Selection Methods

In this section, we define the class of invariance-based feature selection (IBFS) methods through a set of core properties. We then establish their connection to causality using the framework provided by the literature on causal graphical models. Based on this connection, we argue that causal discovery methods offer practical tools for identifying invariant features and can therefore be used as IBFS methods. Finally, we describe the eight causal discovery algorithms employed in our empirical study, which serve as proxies for the IBFS class in our comparative analysis.

Invariance is a key concept here. As noted by [23], it is not new in the literature, it can be traced back to [64], where it appears under the name modularity.

We begin by following Pfister et al. [18] and define an invariant feature set S as follows:

Definition 1 (*Invariant set S (linear case)*)**.** *A set $S \subseteq \{1, \ldots, k\}$, where k is the number of features being considered such that $|S| \leq k$, is called* invariant *with respect to $(Y, \boldsymbol{X})$ if there exist parameters $\mu \in \mathbb{R}$, $\boldsymbol{\beta} \in (\mathbb{R} \setminus \{0\})^{|S|\times 1}$, and $\sigma \in \mathbb{R}_{>0}$ such that:*

(a) For all $t \in \{1, \ldots, T\}$:

$$Y_{t+1} = \mu + \boldsymbol{X}_t^S \boldsymbol{\beta} + \varepsilon_t, \quad \text{with } \varepsilon_t \perp\!\!\!\perp \boldsymbol{X}_t^S,$$

(b) $\varepsilon_1, \ldots, \varepsilon_T \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Here, the index set $\{1, \ldots, k\}$ corresponds one-to-one with the components of the feature vector $\boldsymbol{X}_t := (Y_{t-1}, \ldots, Y_{t-p}, \boldsymbol{Z}_t', \ldots, \boldsymbol{Z}_{t-r}')'$.

While this definition is intuitive and analytically convenient, it assumes a linear and homoskedastic data-generating process. In many practical applications, especially in economics and finance, such assumptions may be overly restrictive. To address this limitation, we now introduce a more general definition that allows for nonlinear relationships and arbitrary noise distributions:

Definition 2 (*Invariant set S (general case)*)**.** *A set $S \subseteq \{1, \ldots, k\}$ is called* invariant *with respect to $(Y, \boldsymbol{X})$ if the conditional distribution of Y_{t+1} given $\boldsymbol{X}_t^S$ remains constant over time, that is,*

$$\mathbb{P}(Y_{t+1} \mid \boldsymbol{X}_t^S) \text{ is the same for all } t \in \{1, \ldots, T\}.$$

A special case of this definition is the environment-based invariance framework [23], where the data is partitioned into distinct environments $e \in \mathcal{E}$ (e.g., experimental conditions, time regimes, or structural breaks), and invariance is required to hold across these environments:

$$\mathbb{P}(Y_{t+1}^{(e)} \mid \boldsymbol{X}_t^{S,(e)}) = \mathbb{P}(Y_{t+1}^{(e')} \mid \boldsymbol{X}_t^{S,(e')}) \quad \text{for all } e, e' \in \mathcal{E}.$$

This formulation captures the core idea that the conditional relationship between Y and a subset of predictors remains stable under distributional changes, aligning with the principle of causal invariance. It also lays the groundwork for applying causal discovery methods under minimal structural assumptions.

A common characteristic among all the models we label as *causal* is their focus on identifying the direct causes of a target variable. This contrasts with feature selection methods that rely solely on predictive performance metrics. In the literature on graphical models, the direct causes of a variable Y_{t+1} are referred to as its *parents*, denoted $\mathrm{PA}(Y_{t+1})$. Let 0 be the index corresponding to Y_{t+1}, and let $\mathcal{I}(\mathrm{PA}(Y_{t+1})) \subseteq \{1, \ldots, k\}$ be the index set corresponding to its parents. Then, under certain assumptions, [23] and [18] show that $S = \mathcal{I}(\mathrm{PA}(Y_{t+1}))$.

Given a set of variables $\mathcal{D}_t = (Y_{t+1}, \boldsymbol{X}_t)$, the objective of causal discovery methods is to recover the underlying causal graph of the system [65]. Implicitly, this means that such methods aim to recover the set $\mathcal{I}(\mathrm{PA}(Y_{t+1}))$. Therefore, we treat causal discovery algorithms as proxies for invariance-based feature selection methods (IBFS).

3.3. Causal Discovery: A Conceptual Paradigm that Encompasses IBFS Methods

Causal discovery refers to the problem of inferring the underlying causal structure among a collection of variables from observed data. More precisely, given a joint distribution over a set of variables, the objective is to recover a causal graph G that encodes the directional relationships governing the data generating process. In contrast to predictive modeling, which is concerned with associations, causal discovery seeks to identify relationships that remain meaningful under interventions and distributional changes. A useful way to formalize this problem is as an inference task mapping the joint distribution $\mathbb{P}(\mathcal{D}_t)$ to a graph G representing the causal data generating mechanism.

In our setting, the relevant variables are given by the target Y_{t+1} together with the components of the feature vector $\boldsymbol{X}_t$ as defined in Section 3.1, where $\boldsymbol{Z}_t$ may include both contemporaneous and lagged exogenous variables. The index set $\{1, \ldots, k\}$ corresponds one to one with the components of $\boldsymbol{X}_t$, so that causal discovery is aimed at identifying which components belong to the parent set of the target, denoted by $\mathrm{PA}(Y_{t+1})$. The parent set is a subset of the full causal graph G, consisting specifically of those nodes with a direct edge into Y_{t+1}. While causal discovery methods generally aim to recover the entire graph G, for the purposes of this paper the relevant output is the local structure $\mathrm{PA}(Y_{t+1}) \subseteq G$, which identifies the predictors whose relationship with the target is structurally stable.

Recovering $\mathrm{PA}(Y_{t+1})$ (through G) from observational data is not possible without additional assumptions. Table 1 summarizes the main identifying assumptions underlying the causal discovery methods considered in this paper. These assumptions restrict the mapping from $\mathbb{P}(\mathcal{D}_t)$ to G, and therefore determine the extent to which causal interpretation is justified. We now discuss the role of each assumption in detail.

Temporal ordering. All methods exploit the time structure of the data by assuming that causes precede their effects. Formally, directed edges are typically restricted to flow from past to future, i.e., from X^i_{t-r} to X^j_t for $r > 0$. This assumption provides a natural source of identification in time series settings and underlies Granger type approaches.

Causal Markov and Faithfulness conditions. These assumptions link the causal graph to the observed distribution. The Causal Markov Condition states that each variable is conditionally independent of its non descendants given its parents, while Faithfulness requires that all conditional independencies in the data are implied by the graph. Together, they allow constraint based methods such as PCMCI to recover (equivalence classes of) causal graphs from conditional independence relations.

Causal sufficiency. This assumption requires that there are no unobserved confounders affecting multiple variables in the system. When it holds, conditional independence relations can be directly attributed to the observed variables. When it fails, some methods (e.g., PCMCI or SeqICP in restricted settings) may still provide valid results, but identification is generally weaker.

Absence of instantaneous effects. Many time series methods assume that causal effects operate only across lags, ruling out contemporaneous relationships between variables at time t. Under this restriction, the causal graph is fully ordered in time and becomes easier to identify. When instantaneous

effects are allowed, additional structural assumptions (e.g., non Gaussianity as in VAR-LiNGAM) are required to determine contemporaneous directions. It is worth noting that methods which exclude instantaneous effects, such as Granger causality and PCMCI, do not achieve structural identification in the econometric sense of structural vector autoregressive (SVAR) models [66], as they do not recover the contemporaneous causal mechanism. The implications of this distinction for our empirical analysis are discussed in Section 3.3.2.

Linearity. Several methods rely on linear structural equations, either as a modeling choice or for statistical tractability. For example, Granger causality and many implementations of SeqICP assume linear relationships between variables. A notable exception is PCMCI, which can accommodate nonlinear dependencies through flexible conditional independence tests [67]. Nonetheless, linearity remains a key practical assumption for most methods considered here.

Gaussian errors. Most of the methods considered in this paper assume Gaussian error terms in their standard formulations, including Granger causality and SeqICP, for which the invariance testing procedure relies on Gaussian residuals [18]. A notable exception is VAR-LiNGAM, which instead requires non Gaussianity of the structural residuals as its primary identifying restriction. By exploiting the statistical independence and non Gaussian distribution of the error terms through independent component analysis, VAR-LiNGAM is able to recover fully directed causal graphs, including contemporaneous causal orderings, without requiring the absence of instantaneous effects [68]. Dynotears and PCMCI, on the other hand, do not impose specific distributional assumptions on the errors in their standard formulations.

Environment invariance. Methods such as Sequential Invariant Causal Prediction (SeqICP) rely on the assumption that the conditional distribution of the target given its direct causes remains invariant across time or across implicit environments. This invariance property replaces traditional assumptions such as Faithfulness and provides an alternative route to identifying causal predictors, even in the presence of certain types of distributional shifts.

Importantly, for most of the methods considered in this paper, the primary objective is not to directly identify invariant predictors in the sense of Definition 3.2. Rather, these methods aim to recover (fully or partially) the causal graph G, from which the parent set $\mathrm{PA}(Y_{t+1})$ can then be extracted. The connection to invariance arises indirectly: under the assumptions summarized in Table 1, the parent set $\mathrm{PA}(Y_{t+1})$ corresponds to a set of predictors whose conditional relationship with the target is stable

across interventions or distributional changes. Consequently, once a causal graph (or an equivalence class thereof) is estimated, invariant predictors can be obtained by selecting the subset of variables identified as direct causes of Y_{t+1}.

From this perspective, invariance is achieved either explicitly, as in methods based on environment invariance such as SeqICP, or implicitly, through structural assumptions that ensure the estimated parent set approximates the set of invariant predictors. Although the individual algorithms differ in implementation and in the specific assumptions they impose, they all seek to approximate the same object: the subset of components of $\boldsymbol{X}_t$ that are causally relevant for predicting Y_{t+1}. Constraint based methods do so through conditional independence testing, Granger type methods exploit temporal predictability, and structural approaches such as VAR-LiNGAM or Dynotears impose additional restrictions to sharpen identification.

For this reason, we interpret the IBFS class as a restricted causal discovery problem, where the goal is not to recover the entire causal graph, but rather to identify a subset of predictors that approximates $\mathcal{I}(\mathrm{PA}(Y_{t+1}))$ and is therefore expected to yield more stable predictions under distributional change.

Method	Time order	Markov / Faith-fulness	Causal suffi-ciency	No inst. effects	Linearity	Gaussian Errors	Env. invari-ance
Pairwise Granger Causality	✓	✓	–	✓	✓	✓	–
Multivariate Granger Causality	✓	✓	✓	✓	✓	✓	–
Sequential Invariant Causal Prediction (SeqICP)	✓	–	–	–	✓	✓	✓
VAR-LiNGAM	✓	–	✓	–	✓	✓	–
Dynotears	✓	–	✓	–	✓	–	–
PCMCI	✓	✓	✓	✓	–	–	–

Table 1: Summary of the main assumptions underlying the causal discovery methods considered in this paper. A checkmark indicates a core identifying restriction in the standard formulation of the method.

3.3.1. *Conditions Under Which IBFS Is Expected to Improve Forecast Robustness*

Having established that IBFS methods aim to approximate the parent set $\mathrm{PA}(Y_{t+1})$, we now address a natural question: under what conditions should forecasts based on invariant predictors outperform those based on purely predictive alternatives?

The theoretical foundation rests on a key result in the causal graphical models literature: under a structural causal model in which no direct intervention acts on the target Y_{t+1}, the conditional distribution $\mathbb{P}(Y_{t+1} \mid \boldsymbol{X}_t^{S^*})$, where $S^* = \mathcal{I}(\mathrm{PA}(Y_{t+1}))$, is invariant across all intervention environments. This has been formally proved by [23] (Proposition 1) and extended to sequential data by [18] (Proposition 6.1 and Theorem 4.1). In contrast, the conditional distribution given an arbitrary predictive set $S \neq S^*$ is generally not invariant. Building on this, [63] introduce the *stable blanket* $\mathrm{SB}_{\mathcal{I}}(Y)$ and prove (Theorem 3.5) that it is both generalizable and regression optimal. They further show (Corollary 3.7) that in linear models, pooled OLS converges to the stable blanket regression only as intervention strength grows without bound, implying that under moderate heterogeneity OLS retains coefficients on unstable predictors and degrades under novel distributional shifts.

Translating these results to our setting, we identify four conditions under which IBFS methods are expected to deliver greater forecast robustness:

1. **Structural invariance of the target equation.** The structural assignment generating Y_{t+1} from its causal parents must remain fixed across regimes, even though predictor distributions may shift. This is plausible when return generating mechanisms are structurally persistent while predictors are subject to policy shocks and crises [18].
2. **Approximate linearity of the data generating process.** The formal results underlying conditions (1) and (4) are derived under a linear Gaussian structural causal model [18, 63]. Our use of macroeconomic predictors at monthly frequency, combined with OLS as the forecasting model, makes the linear approximation a reasonable working assumption. Extensions to nonlinear settings are discussed by [18] (Remark B.1) but lack formal guarantees.
3. **Sufficient heterogeneity in the training sample.** The power of invariance based tests to reject non-invariant sets increases with the magnitude of distributional shifts in the sample [18, Theorems 4.1 and 4.2]. We exploit episodes such as the GFC, the EUR/CHF peg break, and the COVID-19 pandemic as implicit environments providing the requisite variation.
4. **Out-of-sample shifts at least as strong as training period shifts.** The robustness advantage is most pronounced when test period instability matches or exceeds that of the training period. This

follows from Corollary 3.7 of [63]: under weaker shifts, pooled OLS already approximates the stable blanket, whereas under stronger shifts, PBFS methods fitted to training period covariances are most vulnerable to degradation.

Conditions (1), (2), and (3) have formal theoretical support in the results cited above, whereas condition (4) is a hypothesis motivated by the asymptotic analysis and simulation evidence of [63]. A formal extension to the nonstationary, dependent data setting of financial time series is beyond the scope of this paper. Our empirical results in Section 6, where we evaluate IBFS and PBFS methods across crisis and non-crisis periods, provide evidence consistent with these four conditions.

3.3.2. Differences in Structural Identification Across IBFS Methods

Although the IBFS methods considered in this paper all aim to approximate the parent set $\mathrm{PA}(Y_{t+1})$, they differ substantially in their identification mechanisms and in their treatment of contemporaneous causal relationships. Following the terminology of the structural vector autoregressive (SVAR) literature [66], we can organize these differences along two dimensions: (i) whether a method achieves *structural identification* of the causal mechanism, or relies only on *predictive precedence*; and (ii) whether the method permits *contemporaneous causal effects* among the observed variables.

Structural identification versus predictive precedence. Granger causality, in both its pairwise and multivariate formulations, is based on the concept of predictive precedence: a variable X_t^j is said to Granger cause Y_{t+1} if past values of X^j improve the prediction of Y beyond the information contained in the remaining predictors [69]. This notion is fundamentally predictive rather than structural. It does not recover the contemporaneous data generating mechanism and, as emphasized by [66], does not correspond to structural identification in the SVAR sense. In particular, Granger noncausality can fail to detect true causal relationships when instantaneous effects are present, and can falsely detect causal links that arise from omitted contemporaneous confounders.

In contrast, VAR-LiNGAM and Dynotears operate within the SVAR framework and achieve structural identification through additional restrictions on the contemporaneous causal structure. VAR-LiNGAM exploits non-Gaussianity of the structural residuals to recover the full contemporaneous causal ordering via independent component analysis [68], while Dynotears imposes sparsity and acyclicity constraints on the contemporaneous adjacency matrix $\boldsymbol{A}$ through a continuous optimization approach [70]. Both methods thus go beyond predictive precedence and aim to re-

cover the structural form of the data generating process, including instantaneous causal effects.

SeqICP occupies a distinct position. Rather than imposing structural restrictions on the full system, it achieves identification through the principle of *structural invariance*: it searches for predictor sets whose conditional relationship with the target remains stable across implicit time regimes, without requiring knowledge of the regime locations [18]. As discussed in Section 3.2, this invariance based identification does not require the Faithfulness assumption and can accommodate both lagged and contemporaneous causal effects. The identification mechanism is therefore fundamentally different from both the predictive precedence of Granger methods and the distributional restrictions (e.g., non-Gaussianity, sparsity) used by SVAR based approaches.

PCMCI, finally, achieves identification through conditional independence testing under the Causal Markov and Faithfulness assumptions [67]. In its standard formulation, it assumes the absence of contemporaneous causal effects, which simplifies the identification problem by ensuring that the time lag structure fully determines the causal ordering. When this assumption is violated, PCMCI may fail to detect instantaneous causal links or may incorrectly attribute them to lagged relationships.

Treatment of contemporaneous causal effects.. The distinction between methods that permit and those that exclude contemporaneous effects is particularly relevant in financial applications, where variables observed at the same frequency (e.g., monthly macroeconomic indicators) may plausibly influence each other within the observation period. Table 1 summarizes these differences: Granger causality and PCMCI assume the absence of instantaneous effects, whereas VAR-LiNGAM, Dynotears, and SeqICP allow for them, each under different identifying restrictions.

This heterogeneity in identification strategies has practical implications for our empirical analysis. Methods that exclude contemporaneous effects may underestimate the true parent set $\mathrm{PA}(Y_{t+1})$ when instantaneous causal links are present, potentially selecting an incomplete set of invariant predictors. Conversely, methods that allow contemporaneous effects face a harder identification problem and may require stronger assumptions (e.g., non-Gaussianity for VAR-LiNGAM) that may not hold in all financial settings. By including methods from both categories in our empirical comparison, we assess the robustness of our findings to these different identification strategies rather than relying on any single set of structural assumptions.

3.3.3. Multivariate Granger Causality

The multivariate Granger causality test [69, 71] evaluates whether a specific feature $X_t^j \in \boldsymbol{X}_t$, for $j = 1, \ldots, k$, improves the prediction of the target variable Y_{t+1} when included in a vector autoregressive (VAR) model.

This is achieved by comparing two nested predictive models:

$$\text{Full model:} \quad Y_{t+1} = \sum_{\ell=1}^{p} B_\ell \boldsymbol{X}_{t-\ell} + \varepsilon_t,$$

$$\text{Restricted model:} \quad Y_{t+1} = \sum_{\ell=1}^{p} B_\ell \boldsymbol{X}_{t-\ell}^{-j} + \varepsilon'_t,$$

where $\boldsymbol{X}_{t-\ell}$ denotes the vector of lagged regressors (including all p lags of X^j), and $\boldsymbol{X}_{t-\ell}^{-j}$ denotes the same vector with all p lags of the j-th variable removed. The error terms ε_t and ε'_t capture the residuals from each model, respectively.

The Granger non-causality hypothesis is then expressed as:

$$\begin{aligned} H_0 &: X_t^j \text{ does not Granger-cause } Y_{t+1}, \\ &\iff B_\ell^{(j)} = 0 \quad \text{for all } \ell = 1, \ldots, p; \\ H_1 &: \text{At least one } B_\ell^{(j)} \neq 0. \end{aligned}$$

Equivalently, H_0 implies that removing all lags of X_t^j does not significantly increase the model's residual sum of squares (RSS), whereas H_1 implies that including X_t^j improves forecast accuracy by reducing residual variance. This nested hypothesis is typically tested via a Wald or F-test, comparing the residual variances of the full and restricted models:

$$\mathcal{F} = \frac{(RSS_R - RSS_F)/q}{RSS_F/(T-k)},$$

where $q = p$ is the number of excluded coefficients and T is the sample size. Under H_0, this $\mathcal{F}$ statistic follows an F distribution.

When applied iteratively to each candidate predictor X_t^j, this test can be used for causal discovery, identifying variables that exhibit a statistically significant directional influence on Y_{t+1} under the assumption of linear dynamics, no omitted confounders, and covariance-stationary errors.

3.3.4. Sequential Invariant Causal Prediction (SeqICP)

Invariant causal prediction (ICP) [23] and its extension to sequential data, SeqICP [18], form the basis for this method. These approaches identify causal relationships by searching for subsets of predictors for which the conditional distribution of the target variable remains invariant across different environments or time regimes.

The key assumption is causal invariance: there exists a subset of features $S^* \subseteq \{1, \ldots, k\}$ such that the conditional distribution of Y_{t+1} given $\boldsymbol{X}_t^{S^*}$ remains stable across all environments $e, h \in \mathcal{E}$:

$$\mathbb{P}(Y_{t+1}^{(e)} \mid \boldsymbol{X}_t^{S^*,(e)} = x) = \mathbb{P}(Y_{t+1}^{(h)} \mid \boldsymbol{X}_t^{S^*,(h)} = x).$$

This assumption implies that no environment involves a direct intervention on the target variable Y_{t+1} and that there are no unmeasured confounders affecting the relationship between predictors and outcome. SeqICP aims to recover the set S^* of invariant (causal) predictors, thereby enabling predictions that are robust to distributional shifts across environments or time periods.

To test whether a candidate set S is invariant, SeqICP performs a two-step hypothesis testing procedure [18]:

1. Estimate the coefficients $\boldsymbol{\beta}$ in the model

$$Y_{t+1} = \boldsymbol{X}_t^S \boldsymbol{\beta} + \varepsilon_t,$$

   assuming $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and independent of $\boldsymbol{X}_t^S$.
2. Test whether the residuals ε_t are i.i.d. Gaussian with equal variance across environments. This goodness-of-fit test assesses whether the linear relationship is invariant, as required by the null hypothesis:

$$H_{0,S} : \exists \boldsymbol{\beta}, \sigma^2 \text{ such that } \mathbb{P}(Y_{t+1} \mid \boldsymbol{X}_t^S) = \mathcal{N}(\boldsymbol{X}_t^S \boldsymbol{\beta}, \sigma^2).$$

Rejecting $H_{0,S}$ implies that S is not an invariant set, and therefore cannot correspond to the causal parents of Y_{t+1}.

An important clarification concerns the stationarity requirements of SeqICP. The method does not require global stationarity of the multivariate process. On the contrary, as emphasized by [18], it is precisely the non-stationary nature of the system that provides the distributional variation needed to detect causal predictors. What SeqICP requires is that the structural relationship between Y_{t+1} and its causal parents remains invariant, while the marginal distributions of the predictors are allowed to change freely across time.

In our setting, return series are rendered weakly stationary through standard log transformations. However, weak stationarity does not preclude local distributional shifts that SeqICP can exploit. Episodes of heightened volatility, such as the GFC or the COVID-19 pandemic, represent implicit environments in which predictor distributions shift locally while the series remains weakly stationary over the full sample. SeqICP leverages this cross-regime variation to test whether a candidate set S satisfies the invariance condition $H_{0,S}$, distinguishing causal parents from predictors whose relationship with the target is regime dependent. The expanding window scheme used in our forecasting exercise naturally spans multiple volatility regimes, providing the heterogeneity required for identification.

3.3.5. VAR-LiNGAM

The VAR-LiNGAM method, introduced by [68], combines vector autoregression (VAR) with independent component analysis (ICA) to uncover causal relationships in multivariate time series data. For this method, we define an extended variable vector $\boldsymbol{X}_t^* := (Y_t, \boldsymbol{X}_t)$, which includes the target variable and the covariates at time t.

The VAR-LiNGAM model is specified as:

$$\boldsymbol{X}_t^* = \sum_{\ell=1}^{p} A_\ell \boldsymbol{X}_{t-\ell}^* + \boldsymbol{\varepsilon}_t, \tag{1}$$

where $A_\ell \in \mathbb{R}^{(d+1)\times(d+1)}$ is a matrix of autoregressive coefficients at lag ℓ, and $\boldsymbol{\varepsilon}_t$ is a vector of non-Gaussian, mutually independent error terms. The key insight of the method is that ICA can be applied to the VAR residuals to recover independent components, enabling the identification of causal directions under suitable assumptions.

The estimation procedure involves the following steps:

1. Fit a VAR(p) model to the extended vector $\boldsymbol{X}_t^*$ and obtain residuals $\boldsymbol{\varepsilon}_t$ along with coefficient matrices A_ℓ.
2. Apply ICA to the residuals to estimate the mixing matrix and extract statistically independent components.
3. Infer a causal ordering of the variables based on the structure of the independent components.
4. Re-estimate the VAR model using the inferred causal ordering to refine the coefficient matrices A_ℓ.

This approach leverages both the temporal structure and the non-Gaussianity of the data to identify causal relationships in time series. Its success hinges on the assumption of non-Gaussian innovations and the adequacy of the VAR model in capturing the linear temporal dynamics.

3.3.6. Dynotears

Dynotears [70] combines static and dynamic Bayesian network structures to infer causal relationships in time series data. The model captures both contemporaneous associations (static structure) and temporal dependencies (dynamic structure).

Similar to the VAR-LiNGAM model, let $\boldsymbol{X}_t^* := (Y_t, \boldsymbol{X}_t)$ denote the extended variable vector that includes the target and covariates at time t. We stack all T observations of the time series into a data matrix $\boldsymbol{X}^* \in \mathbb{R}^{T\times(d+1)}$. Additionally, we define the lagged data matrix as:

$$\boldsymbol{X}_{\text{lag}}^* := [\boldsymbol{X}_{t-1}^* \mid \cdots \mid \boldsymbol{X}_{t-p}^*] \in \mathbb{R}^{T\times(d+1)p},$$

which collects p lags of the extended variables for each time step.

Dynotears estimates a model by solving the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{A},\boldsymbol{B}} \quad & \frac{1}{2T}\left\|\boldsymbol{X}^* - \boldsymbol{X}^*\boldsymbol{A} - \boldsymbol{X}^*_{\text{lag}}\boldsymbol{B}\right\|_F^2 + \lambda_{\boldsymbol{A}}\|\boldsymbol{A}\|_1 + \lambda_{\boldsymbol{B}}\|\boldsymbol{B}\|_1 \\ \text{subject to} \quad & \boldsymbol{A} \text{ is acyclic,} \end{aligned}$$

where $\boldsymbol{A} \in \mathbb{R}^{(d+1)\times(d+1)}$ encodes contemporaneous (static) relationships at time t, $\boldsymbol{B} \in \mathbb{R}^{(d+1)p\times(d+1)}$ encodes lagged (dynamic) relationships, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_1$ the elementwise norm ℓ_1, $\lambda_{\boldsymbol{A}}, \lambda_{\boldsymbol{B}} > 0$ are regularization parameters that control the sparsity of $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively.

The acyclicity constraint on $\boldsymbol{A}$ ensures that the estimated static graph forms a Directed Acyclic Graph (DAG). The overall objective is to minimize the reconstruction error of the observed time series while enforcing interpretable, sparse, and non-cyclic causal structures across both contemporaneous and temporal dimensions.

3.3.7. PCMCI

PCMCI [67] is a two-stage constraint-based causal discovery algorithm developed for high-dimensional time series data. It extends the PC algorithm [72] to account for temporal dependencies, autocorrelation, and the control of false positive rates.

Let $\boldsymbol{X}^*_t := (Y_t, \boldsymbol{X}_t)$ denote the extended vector that includes both the target and the covariates at time t. PCMCI aims to infer time-lagged causal relationships of the form $X^{j_1}_{t-\tau} \to X^{j_2}_t$, where $j_1, j_2 \in \{0, \ldots, k\}$, by testing for conditional independence between variables at different time lags.

The method proceeds in the following steps:

1. A variant of the PC algorithm (denoted PC_1) is estimates a preliminary set of candidate parents $\hat{\text{PA}}(X^{j_2}_t)$ for each variable $X^{j_2}_t$. This is done by iteratively removing variables that are conditionally independent of $X^{j_2}_t$, given subsets of other variables. A relatively liberal threshold α is typically used in this step to avoid discarding true parents prematurely, thus reducing the dimensionality of the conditional sets in the subsequent step.
2. For each potential causal link $X^{j_1}_{t-\tau} \to X^{j_2}_t$, the algorithm tests the following null hypothesis of conditional independence:

$$H_0 : X^{j_1}_{t-\tau} \perp\!\!\!\perp X^{j_2}_t \mid \hat{\text{PA}}(X^{j_2}_t) \setminus \{X^{j_1}_{t-\tau}\}, \hat{\text{PA}}(X^{j_1}_{t-\tau}).$$

   This Momentary Conditional Independence (MCI) test accounts for both confounding effects and autocorrelation. The test statistic can

be chosen based on the characteristics of the data and includes options such as partial correlation (ParCorr), Gaussian Process Distance Correlation (GPDC), or Conditional Mutual Information (CMI).

3. To determine the statistical significance of each tested link, PCMCI computes a p-value from the MCI test. These p-values can optionally be adjusted using multiple hypothesis correction methods such as the Benjamini-Hochberg procedure, in order to control the false discovery rate.

By combining dimensionality reduction with robust independence testing, PCMCI mitigates the power loss commonly associated with high-dimensional conditioning sets in traditional FullCI (full conditional independence) methods. It has been shown to achieve superior detection rates across a range of linear and nonlinear data-generating processes.

3.4. Prediction-Based Feature Selection Models

In this section, we define the class of prediction-based feature selection (PBFS) methods. We begin by clarifying the key distinction between PBFS and IBFS models, as framed within the literature on causal graphical models. We then relate this distinction to the core objective of the paper: assessing the practical utility of causality-inspired principles, particularly invariance, by comparing them against suitable predictive benchmarks. Building on this motivation, we outline the core properties a method must satisfy to be classified as PBFS. Next, we draw on the taxonomy proposed by [73] to identify which classes of feature selection methods align most closely with these criteria. Finally, we describe the implementation of the selected method.

A central concept in the evaluation of predictive feature selection is the *Markov blanket* of the target variable Y, denoted $\mathrm{MB}(Y)$. In graphical models, $\mathrm{MB}(Y)$ is defined as the minimal set of variables that renders Y conditionally independent of all others in the graph. Formally, if V is the set of all variables, then the Markov blanket satisfies:

$$\forall V_i \in V \setminus (\mathrm{MB}(Y) \cup \{Y\}) : \quad Y \perp\!\!\!\perp V_i \mid \mathrm{MB}(Y).$$

This implies that $\mathrm{MB}(Y)$ is the smallest set of features with maximal predictive power for Y: once $\mathrm{MB}(Y)$ is known, no other variable adds further information about Y. It therefore defines the optimal target set for prediction-based methods under a probabilistic framework.

In contrast to IBFS methods, which aim to identify the *direct causes* of Y—that is, the set $\mathrm{PA}(Y)$—PBFS methods are not constrained by causal interpretability. That is, the goal of finding $\mathrm{MB}(Y)$ is not to orient the edges implied by the graphical structure. Rather, PBFS methods may select variables that correlate with Y even if they are not causally upstream.

These may include *children* of Y, or *spouses* (i.e., other parents of Y's children), which are predictive but not necessarily causal.

Figure 2 illustrates this distinction. The left panel includes all variables in $\mathrm{MB}(Y_t)$—parents, children, and spouses. The right panel shows only the parents $\mathrm{PA}(Y_t)$, which a causal discovery method would aim to recover.

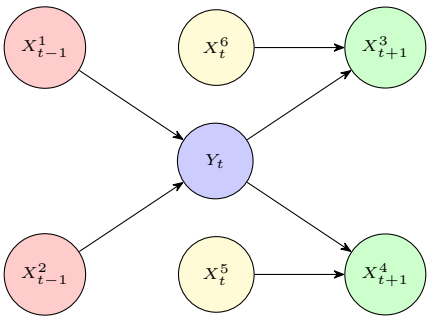
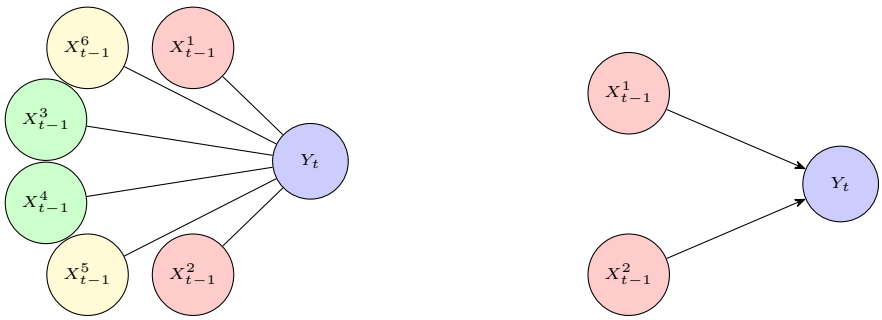


Figure 1: The figure depicts the true causal graph of an underlying time series process. Parents $\mathrm{PA}(Y_t)$ are direct causes. Children and spouses of Y_t complete the Markov blanket $\mathrm{MB}(Y_t)$.



(a) Markov blanket $\mathrm{MB}(Y_t)$ (b) Parents $\mathrm{PA}(Y_t)$ only

Figure 2: The figure depicts a comparison between features selected using the Markov blanket and a selected causal discovery method, provided some assumptions. Parents are direct causes.

The main difference between methods that aim to recover the Markov blanket $\mathrm{MB}(Y)$ and those that target the direct causes $\mathrm{PA}(Y)$ lies in their objectives: the former seeks to maximize in-sample predictive performance, while the latter prioritizes stability across environments. This tension highlights the importance of comparing PBFS and IBFS approaches in settings where the goals of predictive accuracy and causal interpretability may not be simultaneously achievable.

Based on the discussion above, we define a method as PBFS if its primary objective is to select features that explicitly optimize some measure of predictive accuracy. In other words, the focus of the method must be on identifying variables that improve prediction performance, rather than

treating prediction as a secondary outcome of a causal discovery procedure. Moreover, an ideal PBFS method should be model-agnostic, which means that the selection process is not tied to the specific choice of predictive model used to map features to the target. This ensures that any observed performance can be attributed to the feature selection mechanism itself, rather than to its interaction with a particular model class.

Among the main categories of feature selection methods, filters, embedded methods, and wrappers, previous studies concluded that wrapper approaches appear to be the most suitable choice for benchmarking against IBFS. As highlighted by [73], wrapper methods directly optimize predictive performance by using the learning algorithm itself to evaluate candidate subsets, treating it as a black box. This aligns precisely with the goals of PBFS, which prioritize accuracy over invariance. In contrast, filter methods operate independently of the learning algorithm and may overlook subtle interactions between features and the model. Embedded methods, although integrated with specific learners, often constrain the feature selection process to model-specific biases. Within the wrapper family, sequential feature selection (SFS) methods provide a natural and interpretable strategy for benchmarking: they iteratively add (or remove) features based on their incremental contribution to the prediction task, producing a clear ranking of variable importance. Their flexibility and direct alignment with the goal of maximizing predictive power make them an ideal baseline for evaluating the marginal benefits of causality-inspired methods in practical forecasting tasks.

3.4.1. Sequential Feature Selection

Sequential Feature Selection (SFS) is a prediction-based feature selection method and serves as our benchmark for evaluating PBFS approaches. SFS is a stepwise procedure that incrementally adds or removes features from the candidate vector $\boldsymbol{X}_t$ based on their contribution to predictive performance [74]. The method is model-agnostic and can be applied in conjunction with any loss function and prediction model.

In the SFS framework, we evaluate the impact of each candidate feature $X_t^j \in \boldsymbol{X}_t$ by testing whether its inclusion (or exclusion) leads to a statistically significant improvement in the loss. Let $L(Y_{t+1}, \hat{Y}_{t+1|t})$ denote the chosen loss function. The hypotheses are defined as:

$$H_0 : L(Y_{t+1}, \hat{Y}_{t+1|t}) \leq L(Y_{t+1}, \hat{Y}^{-j}_{t+1|t}),$$
$$H_1 : L(Y_{t+1}, \hat{Y}_{t+1|t}) > L(Y_{t+1}, \hat{Y}^{-j}_{t+1|t}),$$

where $\hat{Y}_{t+1|t}$ is the forecast using all selected features, and $\hat{Y}^{-j}_{t+1|t}$ is the forecast excluding X_t^j. Rejecting the null hypothesis supports the inclusion of X_t^j.

The predictive models compared in SFS are of the form:

$$\begin{aligned} \text{Without } X_t^j: \quad & Y_{t+1} = f(\boldsymbol{X}_t^{-j}) + \varepsilon_t, \\ \text{With } X_t^j: \quad & Y_{t+1} = f(\boldsymbol{X}_t) + \varepsilon_t', \end{aligned}$$

where f is a flexible prediction function. In our empirical evaluation, we use both linear regression and random forest models as base learners in the SFS procedure.

SFS can be implemented in two variants: forward and backward selection. In the forward variant, the algorithm starts with an empty feature set and sequentially adds the feature that most reduces the loss, continuing until no further improvement is achieved. In backward selection, the process starts with the full feature set and removes the least relevant feature at each step.

To enforce a purely predictive objective, we adopt the mean squared error (MSE) as the loss function. This choice helps isolate the impact of prediction-optimized feature selection from causality-inspired alternatives such as invariance-based methods.

3.5. Forecasting Model

Once the feature subset is selected, let $\boldsymbol{X}_t^S \in \mathbb{R}^r$ denote the resulting r-dimensional vector of selected features at time t, where $r < n$ and n is the full dimension of $\boldsymbol{X}_t$. The forecasting model maps these features into a prediction of the target:

$$Y_{t+1} = g(\boldsymbol{X}_t^S) + \varepsilon_{t+1},$$

where $g(\cdot)$ is a learned prediction function and ε_{t+1} denotes the residual.

In our analysis, we deliberately adopt a simple linear prediction model to isolate the role of the feature selection mechanism in driving predictive performance. Specifically, we estimate:

$$Y_{t+1} = \boldsymbol{X}_t^S \boldsymbol{\beta} + \varepsilon_{t+1},$$

where the coefficient vector $\boldsymbol{\beta}$ is estimated via ordinary least squares (OLS). This design choice serves a critical function: it ensures that any improvements in predictive accuracy can be attributed to the quality of the selected features, whether derived from IBFS or PBFS, rather than from the complexity of the prediction model itself.

By abstracting away model capacity as a confounding factor, we are better able to assess the central question of this study: whether causality-inspired selection criteria, such as invariance across environments, translate into superior predictive performance in practice.

3.6. Extended Benchmark Forecasting Model

(RC2) As discussed in Section 3, the baseline forecasting model was intentionally kept simple in order to isolate the effects of feature selection. This design choice allows us to attribute any observed gains in predictive performance or stability directly to IBFS and PBFS mechanisms, rather than to increased model complexity. Nevertheless, a natural follow-up question is how these methods perform when compared against more sophisticated forecasting benchmarks, irrespective of whether such models incorporate implicit feature selection mechanisms.

To address this question, we introduce a second set of benchmark models that aim to evaluate the joint performance of IBFS and PBFS methods when paired with more flexible and realistic forecasting frameworks. These extended benchmarks are designed to capture key features of financial time series, such as regime dependence, time variation, and nonlinearities, which are not explicitly modeled in the baseline linear specification.

Before introducing these models, it is worth noting that a natural class of alternatives includes regularization-based linear models, such as Ridge regression and the Least Absolute Shrinkage and Selection Operator (LASSO) [12]. These methods perform shrinkage and, in the case of LASSO, implicit variable selection by penalizing the magnitude of the coefficients. While these approaches are widely used in high-dimensional settings, we deliberately exclude them from the baseline analysis. The reason is that they inherently combine the feature selection and prediction steps into a single optimization problem, making it difficult to disentangle whether any performance gains arise from improved feature selection or from the predictive model itself. For this reason, they are instead incorporated as part of the extended benchmark set.

The first class of models consists of econometric approaches that explicitly account for parameter instability over time. These include regime-switching models, such as the Markov-switching dynamic regression framework of Goldfeld and Quandt [75], in which model parameters evolve according to an unobserved Markov process, allowing for discrete structural breaks. We also consider score-driven or Generalized Autoregressive Score (GAS) models [76], which introduce continuously time-varying parameters updated through the score of the conditional likelihood, enabling smooth adaptation to changing volatility and distributional properties. In addition, we include regularization-based linear models, namely Ridge and LASSO, which address instability through shrinkage and implicit feature selection, providing a parsimonious yet competitive benchmark in high-dimensional environments.

The second class of models includes modern machine learning approaches that address instability through nonlinear modeling and variance reduction. In particular, we consider Random Forests [77], which rely on bootstrap

aggregation (bagging) to reduce variance and improve stability, and Gradient Boosting Machines [78], which sequentially refine predictions by focusing on difficult-to-predict observations. These ensemble methods are well known for their strong empirical performance and robustness in noisy and small-sample environments.

Both econometric and machine learning benchmarks considered here are explicitly designed to handle non-stationarity, regime shifts, and nonlinear dynamics. By comparing IBFS and PBFS approaches against these more demanding alternatives, we assess whether causality-driven feature selection provides incremental value even relative to models specifically engineered to address the same empirical challenges.

3.6.1. Markov-Switching Dynamic Regression

In this extended benchmark, the prediction function is allowed to vary across latent regimes, capturing potential structural changes in the data-generating process.

Specifically, we assume the existence of an unobserved state variable $C_t \in \{1, \ldots, K\}$, which follows a first-order Markov chain with transition probabilities

$$\mathbb{P}(C_{t+1} = j \mid C_t = i) = P_{ij}.$$

Conditional on the regime $C_t = i$, the forecasting model is specified as a regime-dependent linear regression:

$$Y_{t+1} = \mu_i + \boldsymbol{X}_t^S \boldsymbol{\beta}_i + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0, \sigma_i^2),$$

where both the intercept μ_i, the coefficient vector $\boldsymbol{\beta}_i \in \mathbb{R}^r$, and the variance σ_i^2 are allowed to vary across regimes.

In our implementation, the feature vector $\boldsymbol{X}_t^S$ consists of lagged values of the target variable, i.e., $\boldsymbol{X}_t^S = (Y_t, Y_{t-1}, \ldots, Y_{t-p+1})$, so that the model corresponds to a regime-dependent autoregressive specification. The lag order p is either fixed or selected via an information criterion, and the number of regimes K is chosen by minimizing an information criterion over a candidate set.

Given estimates of the model parameters and transition probabilities, forecasting is performed by integrating over the latent regimes. Let $\mathcal{F}_T$ denote the information set available at time T. The regime probabilities are first updated using the transition dynamics:

$$\mathbb{P}(C_{T+1} = j \mid \mathcal{F}_T) = \sum_{i=1}^{K} \mathbb{P}(C_T = i \mid \mathcal{F}_T)\, P_{ij}.$$

Next, the regime-specific conditional forecasts are computed as:

$$\mathbb{E}[Y_{T+1} \mid C_{T+1} = i, \mathcal{F}_T] = \mu_i + \boldsymbol{X}_T^S \boldsymbol{\beta}_i.$$

The final one-step-ahead prediction is then obtained as the expectation over regimes:

$$\mathbb{E}[Y_{T+1} \mid \mathcal{F}_T] = \sum_{i=1}^{K} \mathbb{P}(C_{T+1} = i \mid \mathcal{F}_T) \left(\mu_i + \boldsymbol{X}_T^S \boldsymbol{\beta}_i\right).$$

3.6.2. Score-Driven (GAS) Dynamic Regression

In this extended benchmark, we consider score-driven models, also known as Generalized Autoregressive Score (GAS) models, which allow model parameters to evolve over time as a function of the score of the conditional likelihood. This provides a flexible framework for capturing time variation in predictive relationships, particularly in the presence of changing volatility and heavy-tailed behavior.

Let $\boldsymbol{X}_t^S \in \mathbb{R}^r$ denote the selected feature vector at time t. In contrast to static linear models, the GAS specification allows the coefficient vector to evolve dynamically over time. The forecasting model is specified as:

$$Y_{t+1} = \boldsymbol{X}_t^S \boldsymbol{\beta}_t + \varepsilon_{t+1},$$

where $\boldsymbol{\beta}_t \in \mathbb{R}^r$ is a time-varying parameter vector and ε_{t+1} is a disturbance term whose distribution may exhibit heavy tails.

The key feature of the GAS framework is that the parameter dynamics are driven by the scaled score of the conditional density. Denote by $f(Y_{t+1} \mid \boldsymbol{\beta}_t)$ the conditional density of Y_{t+1}. The score is defined as:

$$\boldsymbol{s}_t = \nabla_{\boldsymbol{\beta}_t} \log f(Y_{t+1} \mid \boldsymbol{\beta}_t),$$

and the parameter update follows the recursion:

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\omega} + \boldsymbol{A}\, \boldsymbol{s}_t + \boldsymbol{B}\, \boldsymbol{\beta}_t,$$

where $\boldsymbol{\omega} \in \mathbb{R}^r$, $\boldsymbol{A} \in \mathbb{R}^{r \times r}$, and $\boldsymbol{B} \in \mathbb{R}^{r \times r}$ are static parameters governing the dynamics.

In our implementation, we assume a Student-t conditional distribution for the disturbances:

$$\varepsilon_{t+1} \sim t_\nu(0, \sigma^2),$$

which enhances robustness to outliers and excess kurtosis commonly observed in financial returns. The score is therefore computed with respect to the Student-t likelihood.

To ensure numerical stability and comparability across time, both the target and the feature vector are standardized using rolling sample moments. Moreover, the feature set is restricted to a small number of predictors (including an intercept and lagged values of the target), reflecting the practical constraints of score-driven estimation in finite samples.

Given the filtered parameter estimate $\boldsymbol{\beta}_T$ at time T, the one-step-ahead forecast is computed as:

$$\mathbb{E}[Y_{T+1} \mid \mathcal{F}_T] = \boldsymbol{X}_T^S \boldsymbol{\beta}_T,$$

where $\mathcal{F}_T$ denotes the information set available at time T.

3.6.3. Regularization-Based Linear Models: Ridge and LASSO

In this extended benchmark, we consider penalized linear models, namely Ridge regression and the Least Absolute Shrinkage and Selection Operator (LASSO), which are standard approaches for handling high-dimensional predictors and mitigating overfitting through regularization.

Let $\boldsymbol{X}_t^S \in \mathbb{R}^r$ denote the selected feature vector at time t. As in the baseline linear specification, we consider a linear forecasting model of the form:

$$Y_{t+1} = \boldsymbol{X}_t^S \boldsymbol{\beta} + \varepsilon_{t+1}.$$

Unlike ordinary least squares, Ridge and LASSO estimate the coefficient vector $\boldsymbol{\beta}$ by solving a penalized optimization problem that shrinks the magnitude of the coefficients.

Ridge Regression. Ridge regression introduces an ℓ_2 penalty on the coefficients, leading to the estimator:

$$\widehat{\boldsymbol{\beta}}^{\text{Ridge}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^r} \left\{ \sum_{t=1}^{T} \left(Y_{t+1} - \boldsymbol{X}_t^S \boldsymbol{\beta} \right)^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right\},$$

where $\lambda > 0$ is a regularization parameter controlling the degree of shrinkage.

The ℓ_2 penalty shrinks coefficients toward zero but does not set them exactly to zero, thereby stabilizing estimation in the presence of multicollinearity and reducing variance, while retaining all predictors in the model.

LASSO. In contrast, LASSO imposes an ℓ_1 penalty, leading to:

$$\widehat{\boldsymbol{\beta}}^{\text{LASSO}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^r} \left\{ \sum_{t=1}^{T} \left(Y_{t+1} - \boldsymbol{X}_t^S \boldsymbol{\beta} \right)^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}.$$

The ℓ_1 penalty induces sparsity in the estimated coefficients, effectively performing variable selection by setting a subset of coefficients exactly to zero. This makes LASSO particularly attractive in high-dimensional environments where many predictors may be irrelevant.

3.6.4. Tree-Based Ensemble Methods: Random Forest and Boosting

In this extended benchmark, we consider nonlinear tree-based ensemble methods, namely Random Forests and Gradient Boosting, which have become standard tools for time series forecasting in high-dimensional environments. These methods are particularly well-suited to capture nonlinear interactions and complex dependencies among predictors, while also providing mechanisms to control overfitting through ensemble learning.

Let $\boldsymbol{X}_t^S \in \mathbb{R}^r$ denote the selected feature vector at time t. In contrast to the baseline linear model, tree-based methods construct a nonlinear prediction function of the form:

$$Y_{t+1} = g(\boldsymbol{X}_t^S) + \varepsilon_{t+1},$$

where $g(\cdot)$ is a flexible function learned from data using an ensemble of regression trees.

In this setting, we define $\boldsymbol{X}_t^S$ to include the full set of macroeconomic predictors together with lags of the target variable, thereby allowing the models to exploit both cross-sectional and temporal dependencies without an explicit parametric restriction.

Random Forest (Bagging). Random Forests build an ensemble of regression trees by repeatedly sampling from the data and averaging the resulting predictions. Formally, let $\{g_b(\cdot)\}_{b=1}^B$ denote a collection of regression trees trained on bootstrap samples of the data. The Random Forest predictor is given by:

$$g_{\mathrm{RF}}(\boldsymbol{X}_t^S) = \frac{1}{B}\sum_{b=1}^{B} g_b(\boldsymbol{X}_t^S).$$

Each tree is grown using a randomized subset of features at each split, which further decorrelates the individual learners. The key effect of this procedure is a substantial reduction in variance through averaging, while maintaining relatively low bias due to the flexibility of individual trees.

Gradient Boosting. In contrast, boosting constructs an additive model sequentially, where each new tree is trained to correct the errors of the current ensemble. Starting from an initial model $g^{(0)}(\cdot)$, the boosting recursion takes the form:

$$g^{(m)}(\boldsymbol{X}_t^S) = g^{(m-1)}(\boldsymbol{X}_t^S) + \eta\, h_m(\boldsymbol{X}_t^S),$$

where $h_m(\cdot)$ is a regression tree fitted to the residuals of the previous iteration and $\eta \in (0, 1]$ is a learning rate controlling the step size.

The final predictor after M iterations is:

$$g_{\mathrm{GB}}(\boldsymbol{X}_t^S) = \sum_{m=1}^{M} \eta\, h_m(\boldsymbol{X}_t^S).$$

Boosting primarily targets bias reduction by sequentially refining the model fit, while regularization through shrinkage and early stopping prevents overfitting.

Bias–Variance Trade-off and Forecasting Implications. From a statistical perspective, Random Forests and Boosting provide complementary approaches to the bias–variance trade-off. Random Forests reduce variance via bagging and randomization, leading to stable predictions in noisy environments. Boosting, on the other hand, focuses on reducing bias by iteratively improving the fit, at the cost of potentially higher variance if not properly regularized.

In the context of time series forecasting, these properties are particularly relevant. Financial and macroeconomic data often exhibit instability, nonlinearities, and low signal-to-noise ratios. Ensemble methods help mitigate these challenges by either stabilizing predictions (Random Forest) or enhancing flexibility (Boosting), making them strong benchmarks when evaluating feature selection methods.

Overall, these models provide a demanding comparison for IBFS and PBFS, as they combine nonlinear approximation with variance-reduction techniques that are known to improve predictive performance in small-sample and high-dimensional settings. A comprehensive overview of these methods in time series applications is provided by [79].

4. Data Description

Our empirical study is based on two distinct forecasting tasks. The first is referred to as the *currency case study*, which involves predicting the returns of the Euro–Swiss franc exchange rate using macroeconomic indicators from both the Eurozone and Switzerland. This task was originally proposed by [18] in the context of causal discovery using the SeqICP method. We extend their analysis by shifting the focus to forecasting performance.

The second task is the *exchange-traded funds (ETFs) case study*, which entails predicting returns of the SPDR S&P 500 ETF (ticker: SPY) based on a broad set of macroeconomic variables obtained from the Federal Reserve Economic Data (FRED) database.

In the following subsections, we describe the datasets and variable construction procedures for both case studies in detail.

4.1. Currency Case Study Dataset

To evaluate the value of causal feature selection methods in a macro-financial setting, we replicate the currency forecasting task proposed by [18]. The objective is to identify monthly macroeconomic predictors that are informative for forecasting the log returns of the Euro–Swiss Franc

(EUR/CHF) exchange rate. While the original study focused on detecting instantaneous causal relationships using the SeqICP framework, we extend the analysis to a forecasting context and evaluate the out-of-sample predictive performance of various feature selection strategies.

The dataset consists of monthly observations from January 1999 to January 2017. The target variable is defined as the log return of the EUR/CHF exchange rate. The predictor set includes both Swiss-specific monetary policy indicators, such as the SNB's call money rate and balance sheet data, and broader macroeconomic indicators reflecting the Swiss economy, such as business confidence and GDP. Additionally, we include macroeconomic indicators for the Euro area, specifically its quarterly GDP, to capture conditions affecting the exchange rate from the Eurozone side.

The macroeconomic data are sourced from three primary institutions: Eurostat, the Organisation for Economic Co-operation and Development (OECD), and the Swiss National Bank (SNB). Table 2 summarizes the data components and their corresponding sources.

Variable Name	Indicator	Description	Source
ldGDPCh	Quarterly GDP (Switzerland)	Log-differenced real GDP growth for Switzerland	Eurostat (2017b)
ldGDPeu	Quarterly GDP (Euro Area)	Log-differenced real GDP growth for the Euro Area	Eurostat (2017a)
ldFCIr	Confidence Index (Switzerland)	Log-differenced business/consumer confidence index	OECD (2017a)
dCPI	Monthly CPI (Switzerland)	Monthly inflation rate (first difference of CPI)	OECD (2017b)
ldRIMFr, ldMAr, ldCHFr, ldtotOAr	Monthly Balance Sheet Data (SNB)	Log-differenced SNB balance sheet components (reserves, assets, FX positions)	SNB (2017)
dCMR	Monthly Call Money Rate (SNB)	Monthly change in short-term interest rate	SNB (2017)
ldEXME	Monthly CHF Average Exchange Rate	Log-differenced CHF nominal effective exchange rate	SNB (2017)

Table 2: Data sources for EUR/CHF forecasting task (January 1999 to January 2017)

All time series are preprocessed according to the transformations described in Table 2 of [18] to ensure stationarity and comparability across variables.

4.2. Exchange-traded Funds (ETFs) Case Study Dataset

To assess the practical value of feature selection methods in a high-dimensional macro-financial forecasting context, we construct a dataset aimed at predicting the log returns of the SPDR S&P 500 ETF (SPY). This ETF tracks the performance of the S&P 500 index and is widely used as a proxy for the overall U.S. equity market. The objective of this task is to identify macroeconomic predictors that are informative for forecasting monthly SPY returns.

The dataset consists of monthly observations from February 2000 to April 2023. The target variable is defined as the log return of the monthly closing price of SPY, with price data obtained from Bloomberg. The predictor set is constructed from a comprehensive panel of U.S. macroeconomic indicators extracted from the FRED-MD database curated by the Federal Reserve Bank of St. Louis [24].

FRED-MD includes a wide array of economic time series organized into eight categories. We retain six of these categories, excluding Group 6 (Interest Rate and Exchange Rates) and Group 8 (Stock Market), since these contain financial asset prices that may embed contemporaneous or forward looking information about equity returns, potentially introducing look ahead bias into the feature selection and forecasting process. Lags of the SPY return series are included separately as part of the feature set, as noted above.

The remaining predictor set consists of the following groups: Group 1 (Output and Income), containing 17 series capturing industrial production and personal income; Group 2 (Labor Market), including 32 series covering employment, unemployment, and weekly hours; Group 3 (Consumption and Orders), containing 10 series on housing starts and building permits; Group 4 (Orders and Inventories), including 14 series on manufacturing orders, retail sales, and business inventories; Group 5 (Money and Credit), comprising 14 series on monetary aggregates, reserves, and consumer credit; and Group 7 (Prices), containing 21 series on producer and consumer price indices.

Each series is accompanied by metadata that specifies recommended transformations to ensure stationarity, as well as groupings and descriptions for interpretability. These transformation codes are applied as per the guidelines provided in the FRED-MD documentation, and the resulting features are standardized prior to modeling.

This predictor set is well established in the macroeconomic forecasting literature and has been extensively used in studies of return predictability

and macroeconomic risk [24, 80, 81, 82]. By restricting the panel to real economic activity, labor market, and price variables, we ensure that the predictors reflect fundamental macroeconomic conditions rather than contemporaneous financial market signals, providing a cleaner test of whether causality inspired feature selection can identify economically meaningful predictors of equity returns. In Section 6, we examine whether the variables selected by the IBFS and PBFS methods are economically interpretable and consistent with established findings in the return prediction literature.

5. Experimental Setup

This section outlines the experimental pipeline used to evaluate feature selection methods across two forecasting tasks: the ETF and the currency case studies. We begin by describing the preprocessing steps applied to the target return series and the corresponding macroeconomic predictor datasets for each case.

Next, we introduce the performance metrics used to evaluate forecasting accuracy and assess the predictive value of the selected features. Finally, we describe the complete training procedure, which consists of three stages:

1. Optional feature filtering to reduce dimensionality when appropriate,
2. Application of the feature selection algorithms under evaluation,
3. Forecasting using the resulting selected feature set.

5.1. Data Pre-processing

This section describes how we prepare the data for both forecasting tasks: predicting SPY ETF returns and Euro–Swiss Franc exchange rate movements. In each case, our goal is to align the target return series with macroeconomic predictors, ensuring consistency in frequency, timing, and scale.

For the ETF case study, we begin with daily SPY closing prices, from which we compute monthly log returns based on end-of-month values. These returns are then merged with macroeconomic indicators from the FRED-MD dataset. Because the FRED-MD database already accounts for release lags and is organized with research-ready vintages, our preprocessing focuses on applying the recommended transformation codes to each time series to ensure stationarity. The transformed indicators are then standardized, and the combined dataset is restricted to months with complete observations across all selected predictors.

For the currency case study, the target is defined as the monthly log return of the EUR/CHF exchange rate. The predictor set includes three macroeconomic indicators that provide broad information about the economic environment in Switzerland and the Euro area. Specifically, we include the log returns of interpolated Swiss GDP (converted to monthly

frequency and adjusted for exchange rate effects), the log returns of interpolated Eurozone GDP, and the Swiss inflation rate derived from monthly changes in the consumer price index (CPI). These variables were selected for their relevance to macroeconomic fundamentals rather than central bank balance sheet composition, aligning the dataset with our forecasting emphasis. All time series are processed to ensure stationarity where appropriate and standardized prior to modeling. The resulting datasets are fully aligned in frequency and scale, enabling a consistent evaluation of feature selection methods across both tasks.

5.2. Estimation and Recursive Prediction Procedure

The evaluation of feature selection methods is conducted using a recursive estimation and prediction procedure. We adopt an expanding-window approach, beginning with an initial training window of seven years of monthly data. At each iteration, the window is expanded by one additional month, and predictions are generated in a rolling, out-of-sample fashion.

Each iteration of the evaluation consists of two steps. The first step applies feature selection algorithms to the training data, and the second step uses the selected features to make a one-step-ahead forecast.

In the feature selection step, we augment the feature matrix by including one lag of the target return series (SPY or EUR/CHF) and up to three lags of each macroeconomic indicator. The resulting lagged dataset is passed to the feature selection algorithms described in Sections 3.2 and 3.4, which return a subset of selected features for the current training window.

In the prediction step, we fit a linear regression model via ordinary least-squares using only the selected features, as described in Section 3.5. The model generates a forecast for the next month. This prediction is stored, the training window is expanded to include the newly observed data point, and the procedure is repeated until the end of the sample period.

5.3. Performance Evaluation

To address the central questions of this study, we evaluated the predictive performance of causality-inspired feature selection methods, particularly those grounded in the principle of invariance, compared to conventional predictive and data-driven approaches. Our objective is not only to measure accuracy in absolute terms, but, more importantly, to assess the relative performance across methods in a way that isolates the added value of causality-inspired selection criteria.

This comparative perspective is motivated by an ongoing debate about the practical value of causal reasoning in forecasting applications [22]. Although some recent work has questioned whether causal inference improves prediction, our aim is to contribute to this research investigation through a

carefully structured empirical analysis. In particular, we assess the downstream utility of selected features by measuring their contribution to predictive accuracy and economic performance.

To evaluate predictive accuracy, we report two key metrics: the root mean squared error (RMSE) and a modified version of the R^2 statistic. The latter replaces the unconditional mean with a random walk benchmark, which is more appropriate for return forecasting tasks where signal-to-noise ratios tend to be low [34]. These metrics are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(Y_{t+1} - \hat{Y}_{t+1|t})^2},$$

$$\text{Modified } R^2 = 1 - \frac{\sum_{t=1}^{T}(Y_{t+1} - \hat{Y}_{t+1|t})^2}{\sum_{t=1}^{T}(Y_{t+1} - Y_t)^2},$$

where $\hat{Y}_{t+1|t}$ denotes the forecast made at time t, and Y_t serves as the lagged benchmark.

Beyond traditional accuracy metrics, we evaluate the economic relevance of forecasts by implementing a simple trading strategy and computing standard portfolio performance measures. At the end of each month, we take a long position if the predicted return is positive and a short position otherwise. The monthly return of this strategy is evaluated as follows:

$$R_t^{\text{strat}} = \text{sgn}(\hat{Y}_{t+1|t}) \cdot Y_{t+1},$$

where $\text{sgn}(\cdot)$ denotes the sign function.

We assess the risk-adjusted performance of each strategy using the Sharpe and Sortino ratios, defined respectively as:

$$\text{Sharpe Ratio} = \frac{\mathbb{E}[R_t^{\text{strat}}]}{\sigma(R_t^{\text{strat}})},$$

$$\text{Sortino Ratio} = \frac{\mathbb{E}[R_t^{\text{strat}}]}{\sigma^-(R_t^{\text{strat}})},$$

where $\sigma(\cdot)$ is the standard deviation of strategy returns and $\sigma^-(\cdot)$ is the downside deviation, computed as:

$$\sigma^-(R_t^{\text{strat}}) = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\min(0, R_t^{\text{strat}})^2}.$$

Crisis regime classification. To partition the out-of-sample period into crisis and normal regimes, we construct a realized volatility measure based on a rolling window of the target return series. Specifically, we define the annualized rolling volatility at time t as:

$$\hat{\sigma}_t = \sqrt{12} \cdot \hat{s}_t, \quad \text{where} \quad \hat{s}_t = \sqrt{\frac{1}{w-1} \sum_{j=0}^{w-1} \left(Y_{t-j} - \bar{Y}_{t|w}\right)^2}$$

is the rolling sample standard deviation computed over a window of $w = 6$ months, and $\bar{Y}_{t,w} = \frac{1}{w} \sum_{j=0}^{w-1} Y_{t-j}$ is the corresponding rolling mean. The crisis indicator variable is then defined as:

$$\mathbb{I}_t^{\text{crisis}} = \begin{cases} 1 & \text{if } \hat{\sigma}_t \geq \bar{\sigma}, \\ 0 & \text{otherwise,} \end{cases}$$

where $\bar{\sigma} = \frac{1}{T} \sum_{t=1}^{T} \hat{\sigma}_t$ is the full sample mean of the annualized rolling volatility series. The sets of crisis and normal time indices are then given by $\mathcal{T}_{\text{crisis}} = \{t : \mathbb{I}_t^{\text{crisis}} = 1\}$ and $\mathcal{T}_{\text{normal}} = \{t : \mathbb{I}_t^{\text{crisis}} = 0\}$, respectively. This classification labels as crisis any month in which realized volatility exceeds its historical average, providing a data driven and asset specific regime partition that does not rely on ex post event dating. The use of the sample mean as the threshold ensures that the crisis label captures periods of above average market stress, which naturally encompass episodes such as the Global Financial Crisis, the EUR/CHF peg break, and the COVID-19 pandemic, without requiring prior knowledge of their timing.

Gap definition. In our empirical analysis, we report all four metrics both in their actual levels and as *gap* measures that quantify the difference in performance between crisis and normal periods. For any performance metric $M \in \{\text{RMSE}, R^2, \text{Sharpe}, \text{Sortino}\}$, we define the corresponding gap as:

$$\Delta M = M_{\text{crisis}} - M_{\text{normal}},$$

where M_{crisis} and M_{normal} denote the metric computed over the crisis and normal subsamples, respectively. For the RMSE gap, a negative value indicates that the method achieves lower prediction error during crises than during normal periods, while for the R^2, Sharpe, and Sortino gaps, a positive value indicates superior performance during crises. The gap measures are particularly informative for our research question, as they directly capture the extent to which a given feature selection method maintains or improves its performance under distributional stress. A method that exhibits a smaller deterioration in performance during crisis periods, reflected in a gap closer to zero or positive, provides evidence of greater forecast robustness.

In all cases, we focus on relative comparisons. Our main interest lies in whether methods guided by causal principles yield better predictive performance and more favorable risk-return profiles compared to purely predictive benchmarks. This dual evaluation, combining statistical accuracy with economic impact, offers a comprehensive view of the practical utility of each method.

6. Empirical Results

This section presents the empirical findings, focusing on the relative performance of IBFS and PBFS feature selection methods. We evaluate results along two dimensions: predictive accuracy and economic performance. To assess the robustness of the features selected by each method, we partition the out-of-sample period into crisis and normal regimes using the volatility based classification defined in Section 5.3. As described therein, a month is classified as a crisis if the annualized rolling volatility $\hat{\sigma}_t$ exceeds the full sample mean $\bar{\sigma}$. Our working hypothesis is that IBFS methods, grounded in causal invariance principles, are better suited to select features that yield more robust and stable predictions during such shifts.

6.1. Prediction accuracy

Currency Case Study: We evaluate the performance of causal and non-causal methods using the same currency dataset as [18]. Table 3 reports the out-of-sample R^2, RMSE, and the Generalization Gap for each feature selection model. The Generalization Gap is defined as the R^2 or RMSE difference between the crisis minus the normal period. We report both the actual levels of each metric across regimes and the corresponding gaps. The absolute levels are included for transparency, while the gaps serve as the primary basis for comparing robustness and stability across methods, as discussed in Section 3.

Across the PBFS baselines (top block of Table 3), predictive accuracy is heterogeneous and markedly sensitive to the crisis regime. Sequential Forward Selection with standard K-fold CV (SFSCV) yields out-of-sample R^2 in the -0.28–0.29 range during normal periods and 0.31–0.45 during crises, with RMSE between 0.62 and 0.84 in normal periods, rising to 2.40–2.71 during crises. The best performing PBFS variant in terms of robustness is SFSCV B SVM, which achieves the highest normal period R^2 (0.29) and lowest normal period RMSE (0.62) among PBFS methods, with a moderate R^2 gap of 0.13. However, SFSCV F Lin achieves the smallest RMSE gap (1.56) and highest crisis R^2 (0.45), though its normal period R^2 is negative (-0.28), indicating poor baseline performance. Substituting the usual K-fold split by a rolling Time Series CV (SFSTSCV) does not mitigate fragility: R^2 gaps range from 0.25 to 0.55 and RMSE gaps from 1.77

to 1.95. Nonlinear learners (RF, SVM) behave similarly, indicating that the main limitation lies in the instability of the selected feature sets rather than in model capacity.

By contrast, the IBFS methods (bottom block) display both higher accuracy and striking robustness and stability across regimes. VAR-LiNGAM achieves the best overall performance, with R^2 of 0.50 during normal periods and 0.59 during crises, the lowest RMSE in both regimes (0.52 and 2.05), and the smallest R^2 gap (0.09) and RMSE gap (1.53) among all IBFS methods. SeqICP and PCMCI perform comparably, both reaching R^2 of 0.59 during crises and RMSE gaps of 1.53 and 1.52, respectively. Dynotears also delivers strong results, with the smallest R^2 gap among all methods (0.05), though its crisis RMSE (2.16) is slightly higher. Even the weakest IBFS method (Pairwise Granger, RMSE gap of 1.71) outperforms most PBFS baselines in terms of robustness and stability.

Comparing across groups, the IBFS methods reduce normal period RMSE by approximately 22%–37% relative to the strongest PBFS baseline (from 0.62–0.84 for PBFS to 0.52–0.59 for IBFS) and consistently achieve higher R^2 in both regimes. Crucially, the IBFS RMSE gaps (1.52–1.71) are uniformly smaller than those of the PBFS methods (1.56–1.95), confirming that the predictive advantage of causal feature selection is preserved and even amplified under distributional stress.

Model	R^2 OOS		RMSE		Generalization Gap	
	Normal	Crisis	Normal	Crisis	R2 Gap	RMSE Gap
–SFSCV B Lin	-0.27	0.31	0.83	2.71	0.58	1.88
SFSCV B RF	0.01	0.43	0.74	2.44	0.42	1.70
–SFSCV B SVM	**0.29**	0.42	**0.62**	2.47	**0.13**	1.85
–SFSCV F Lin	-0.28	**0.45**	0.84	**2.40**	0.73	**1.56**
SFSCV F RF	-0.12	0.38	0.78	2.54	0.50	1.76
–SFSCV F SVM	-0.12	0.35	0.78	2.60	0.47	1.82
–SFSTSCV B Lin	0.06	0.31	0.72	2.67	0.25	1.95
SFSTSCV B RF	0.01	0.40	0.74	2.51	0.39	1.77
SFSTSCV B SVM	-0.14	0.37	0.79	2.58	0.51	1.79
SFSTSCV F Lin	-0.21	0.34	0.81	2.61	0.55	1.80
SFSTSCV F RF	-0.06	0.35	0.76	2.59	0.40	1.83
SFSTSCV F SVM	-0.07	0.37	0.77	2.56	0.44	1.80
SeqICP	0.47	**0.59**	0.54	2.07	0.11	1.53
VAR-LiNGAM	**0.50**	**0.59**	**0.52**	**2.05**	0.09	1.53
Dynotears	**0.50**	0.55	**0.52**	2.16	**0.05**	1.63
Multivariate Granger	0.37	0.52	0.59	2.21	0.15	1.63
PCMCI	0.47	**0.59**	0.54	**2.06**	0.12	**1.52**
Pairwise Granger	0.39	0.49	0.58	2.29	0.11	1.71

Table 3: This table reports out-of-sample R^2 and RMSE for the non-crisis ("normal") and crisis periods in the currency case study. The last two columns show the generalization gap for each metric, calculated as the difference between performance in crisis and normal periods. Higher R^2 values and lower RMSE indicate better predictive performance. For the generalization gap, a smaller RMSE increase (lower RMSE gap) and a smaller decline in R^2 (or positive gap) reflect better robustness to distributional shifts. The best-performing model for each group (PBFS and IBFS) and the overall best models are highlighted in bold.

Figures 3 and 4 complement the aggregate evidence by tracing a rolling window RMSE for EUR/CHF forecasts. Each figure presents the rolling RMSE levels in the top panel, with the corresponding RMSE differences relative to a PBFS benchmark in the bottom panel (SFSCV F Lin in Figure 3 and SFSCV B SVM in Figure 4). Shaded regions denote crisis periods identified by the volatility based regime classification described in Section 5.3.

Across both figures, several distinct regimes emerge. During tranquil periods, all methods exhibit relatively similar performance, with low and tightly clustered RMSE values. However, as market conditions deteriorate, differences across methods become more pronounced. Three volatility regimes are clearly discernible: a tranquil pre-2008 phase, an extended turmoil spanning the Global Financial Crisis, the Euro area debt episode and the 2011–13 SNB floor, and the sharp dislocation surrounding the January 2015 removal of that floor. Across the entire horizon, the four IBFS methods (SeqICP, VAR-LiNGAM, Dynotears, and PCMCI) tend to maintain lower RMSE relative to the PBFS benchmarks during crisis periods. This is evident in the top panels, where the PBFS methods show sharper increases in RMSE during stress episodes, while IBFS methods rise more

moderately. The effect is especially visible during the 2008–2009 crisis and the 2011–2012 period, where the gap between PBFS and IBFS widens substantially.

The bottom panels quantify this advantage. Positive values indicate that the IBFS model outperforms the PBFS benchmark; the differences are consistently positive and often large during crisis periods, reaching peaks above 0.6–0.8 in several episodes. In contrast, during more stable periods, the differences shrink toward zero and occasionally become slightly negative, suggesting that the advantage of IBFS methods is primarily concentrated in periods of heightened volatility. Importantly, even after the extreme regime shift in early 2015, the causal approaches preserve a 0.3–0.4 RMSE edge while the PBFS baselines deteriorate markedly.

Comparing the two figures, the results are robust across both the linear (SFSCV F Lin) and nonlinear (SFSCV B SVM) PBFS benchmarks. While the magnitude of the differences varies, the qualitative pattern remains the same: causality inspired feature selection methods deliver more robust and stable predictive performance under distributional shifts.

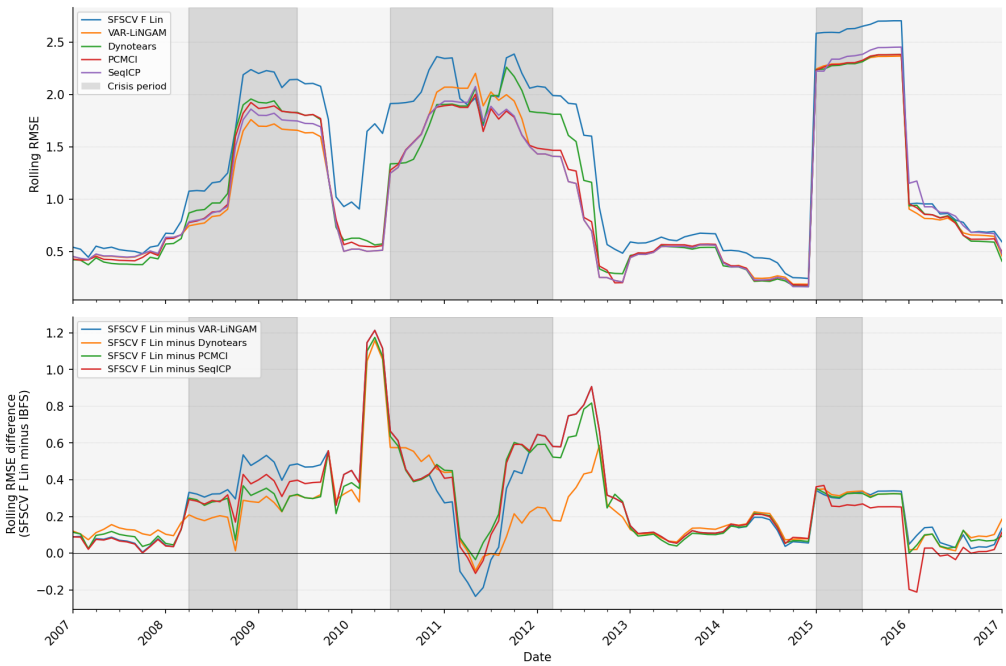


Figure 3: The figure shows the rolling RMSE for EUR/CHF return forecasts using different feature selection methods: SFSCV F Lin (PBFS benchmark), VAR-LiNGAM, Dynotears, PCMCI, and SeqICP (top panel). The bottom panel reports the RMSE differences between the PBFS benchmark (SFSCV F Lin) and each IBFS method. Positive values indicate superior performance of the IBFS methods relative to the PBFS benchmark. Shaded regions denote crisis periods.

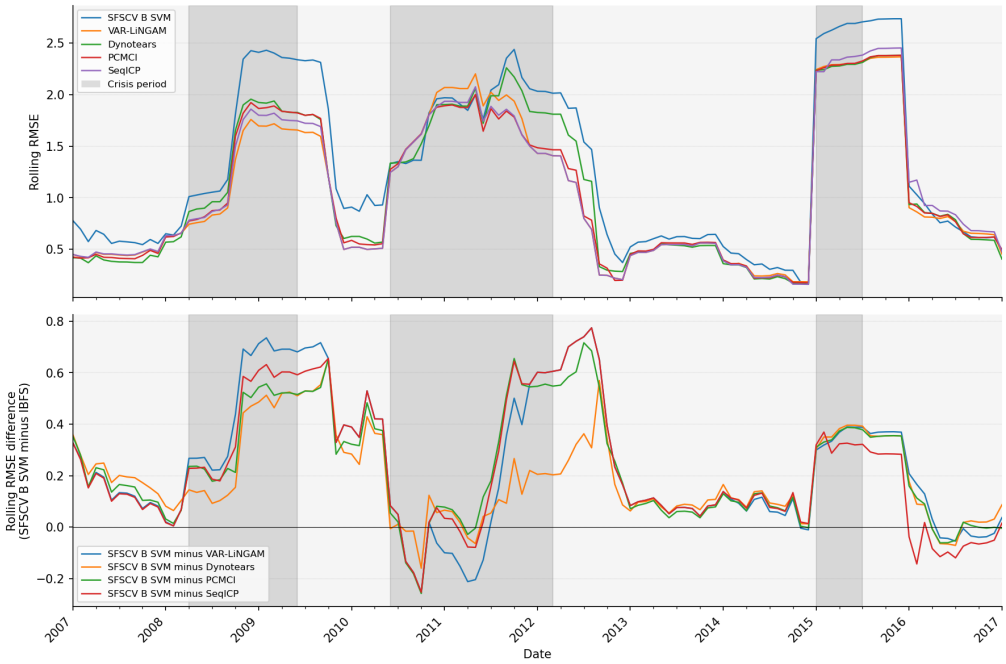


Figure 4: The figure shows the rolling RMSE for EUR/CHF return forecasts using different feature selection methods: SFSCV B SVM (PBFS benchmark), VAR-LiNGAM, Dynotears, PCMCI, and SeqICP (top panel). The bottom panel reports the RMSE differences between the PBFS benchmark (SFSCV B SVM) and each IBFS method. Positive values indicate superior performance of the IBFS methods relative to the PBFS benchmark. Shaded regions denote crisis periods.

The directed graphs in Figure 5 illustrate the causal relationships inferred by six different discovery algorithms for predicting monthly EUR/CHF log returns (ldEXME). The most consistently selected predictors are the log-differenced Swiss GDP (ldGDPCh) and the first lag of the target variable itself (ldEXME), each appearing in three out of six directed graphs. This pattern is consistent with findings from [18], which emphasize the relevance of real economic activity in exchange rate analysis.

The next most frequently identified variables are balance sheet–related indicators, including the log change in SNB-held Swiss franc assets (ldCHFr) and the Swiss call money rate (dCMR), both of which appear in two of the six graphs. These features reflect the role of monetary policy and central bank operations in shaping currency dynamics, albeit with less universal agreement across methods. Other variables, such as consumer confidence (ldFCIr) and inflation (dCPI), are identified more sporadically, suggesting their influence may be more conditional or model-dependent.

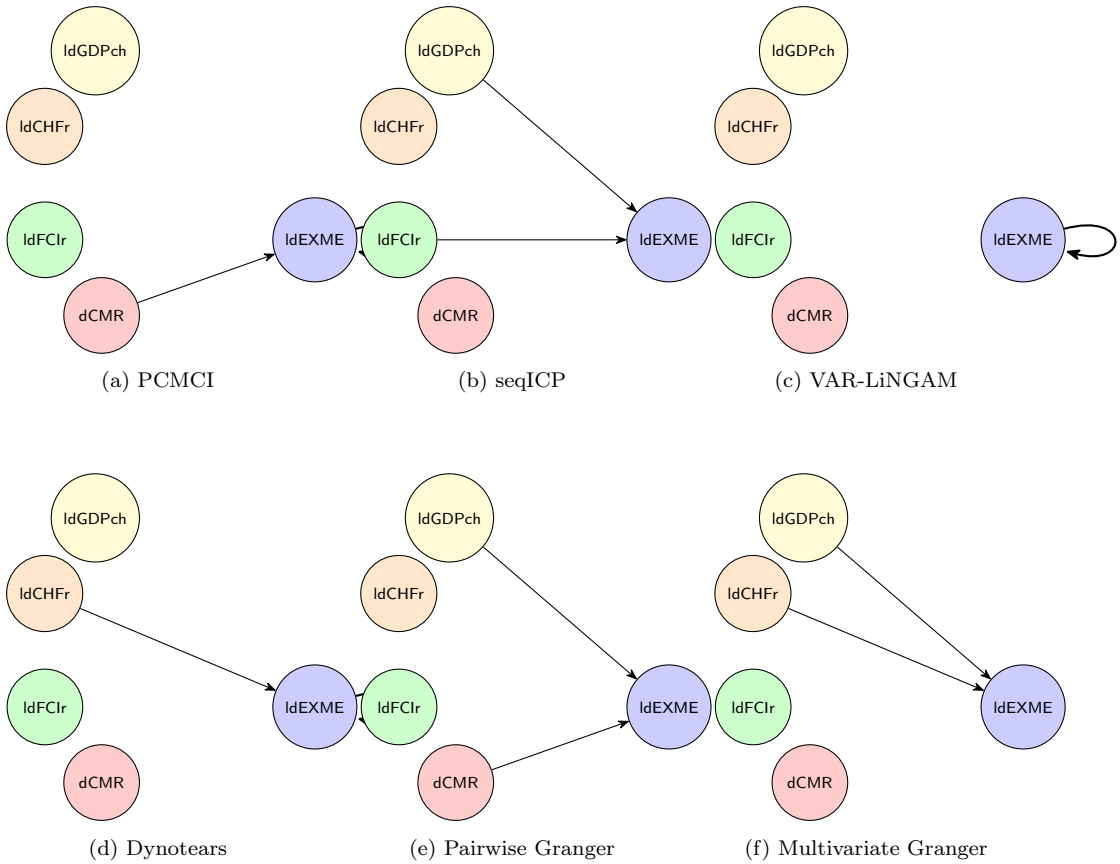


Figure 5: The figure presents a summary graph comparing the output of various causal discovery methods applied to the currency case study at the last time step of the recursive out-of-sample exercise. Each subfigure shows the directed edges inferred by a specific method, where nodes represent candidate predictors and edges indicate discovered causal relationships toward the target variable `ldEXME` (log return of the EUR/CHF exchange rate). Self-loops represent identified autocorrelation. Methods include PCMCI (a), seqICP (b), VAR-LiNGAM (c), Dynotears (d), Pairwise Granger (e), and Multivariate Granger (f).

Exchange-traded Funds (ETFs) Case Study: Table 4 summarizes the out-of-sample R^2, RMSE, and generalization gap metrics for the ETF forecasting task across causal (IBFS) and non-causal (PBFS) feature selection methods. We report both the actual levels across regimes and the corresponding gaps, with the gaps serving as the primary basis for comparing robustness and stability across methods. Unlike the currency case study, where we tested the full range of PBFS methods, here we apply only SFSCV B SVM as the PBFS representative, as it was arguably the best performing PBFS variant in the currency experiment, and re-running all PBFS benchmarks on the larger ETF dataset would introduce substantial computational costs without adding proportional insight.

Among the IBFS methods, SeqICP is excluded from this case study because it returns the empty set for the selected features at every time

step of the expanding window estimation, effectively producing no causal predictors throughout the evaluation period. This is consistent with the conservative nature of the method, which outputs the empty set when the invariance assumption cannot be confidently verified, as discussed in Section 3.2.

The PBFS benchmark (SFSCV B SVM) exhibits severely negative R^2 values in both normal (-13.52) and crisis (-0.76) periods, indicating that it performs substantially worse than a naive random walk forecast. While its R^2 gap of 12.76 and negative RMSE gap of -0.04 might superficially suggest improvement during crises, these figures must be interpreted with caution: the model starts from such a poor predictive base that any relative improvement is economically meaningless. The RMSE values (0.16 in normal periods and 0.12 in crises) are also considerably higher than those of the IBFS alternatives.

The IBFS methods display markedly superior predictive performance in both regimes. VAR-LiNGAM achieves the best overall results, with R^2 of 0.51 during normal periods and 0.52 during crises, the lowest RMSE in both regimes (0.03 and 0.06), and the smallest R^2 gap (0.01) and RMSE gap (0.03) among all methods. This near zero R^2 gap indicates that its predictive accuracy is essentially unchanged across regimes, providing strong evidence of robust and stable performance under distributional shifts. Dynotears and PCMCI perform comparably in normal periods (R^2 of 0.51), though both experience modest declines during crises (R^2 of 0.49 and 0.46, respectively), with RMSE gaps of 0.04. Multivariate Granger shows a more pronounced deterioration (R^2 gap of -0.20), while Pairwise Granger exhibits the largest decline among IBFS methods (R^2 gap of -0.43), suggesting that its simpler identification strategy is less resilient to the distributional shifts present in the ETF dataset.

Model	R^2 OOS		RMSE		Generalization Gap	
	Normal	Crisis	Normal	Crisis	R2 Gap	RMSE Gap
SFSCV B SVM	**-13.52**	**-0.76**	**0.16**	**0.12**	**12.76**	**-0.04**
VAR-LiNGAM	**0.51**	**0.52**	**0.03**	**0.06**	**0.01**	**0.03**
Dynotears	**0.51**	0.49	**0.03**	**0.06**	-0.01	0.04
Multivariate Granger	0.49	0.29	**0.03**	0.08	-0.20	0.05
PCMCI	**0.51**	0.46	**0.03**	0.07	-0.06	0.04
Pairwise Granger	0.46	0.03	**0.03**	0.09	-0.43	0.06

Table 4: This table reports out-of-sample R^2 and RMSE for the non-crisis ("normal") and crisis periods in the ETF case study. The last two columns show the generalization gap for each metric, calculated as the difference between performance in crisis and normal periods. Higher R^2 values and lower RMSE indicate better predictive performance. For the generalization gap, a smaller RMSE increase (lower RMSE gap) and a smaller decline in R^2 (or positive gap) reflect better robustness and stability under distributional shifts. The best-performing model for each group (PBFS and IBFS) and the overall best models are highlighted in bold.

Figure 6 complements the ETF forecasting analysis by showing rolling window RMSEs for SPY return predictions across five feature selection methods. Shaded regions denote crisis periods identified by the volatility based regime classification. The top panel shows that, in normal market conditions, all methods deliver similar predictive performance, with low and tightly clustered RMSE values. However, during periods of heightened volatility, the PBFS benchmark (SFSCV B SVM) exhibits more pronounced spikes in prediction error. This is particularly evident around 2008–2009 during the Global Financial Crisis and during the 2020–2021 period, where SFSCV B SVM shows sharp increases in RMSE, while VAR-LiNGAM, Dynotears, and PCMCI remain comparatively more robust and stable.

The bottom panel reports the RMSE differences between the PBFS benchmark and each IBFS method. Positive values indicate superior performance of the causal methods. During crisis periods, these differences become significantly positive, reaching levels above 0.4 in the early sample and around 0.15–0.2 in more recent episodes. In contrast, during tranquil periods, the differences are close to zero, indicating similar performance across methods. This temporal pattern reinforces the evidence presented in Table 4: while the PBFS benchmark suffers from severe predictive degradation under stress, IBFS methods maintain their accuracy and demonstrate significantly enhanced robustness and stability across regimes. The concentration of the IBFS advantage in crisis periods further supports the central hypothesis that invariant predictors yield improved forecast resilience precisely when it matters most, during severe distributional shifts.

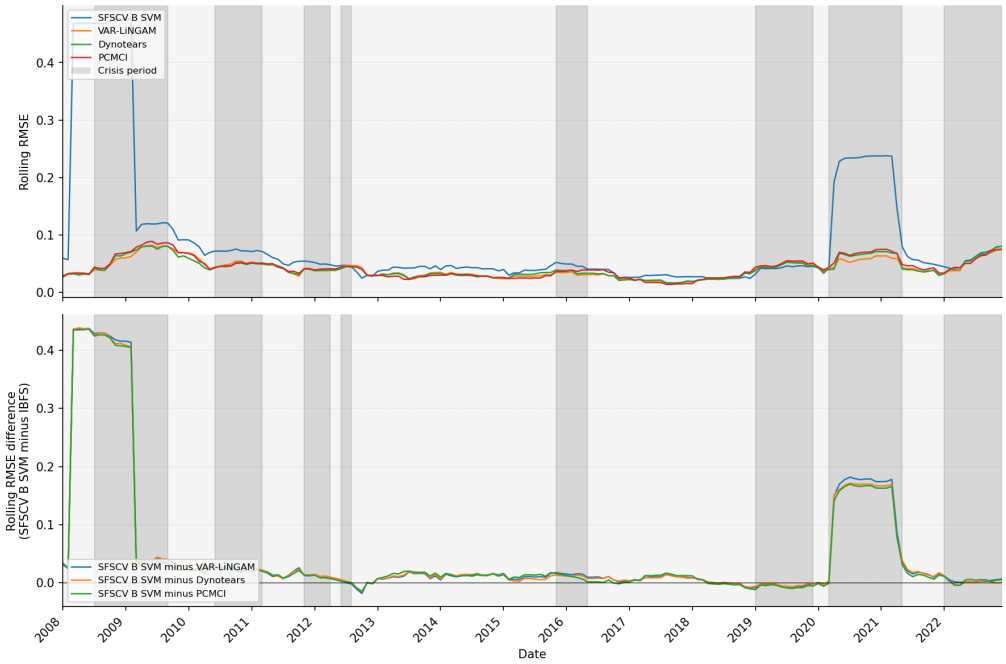
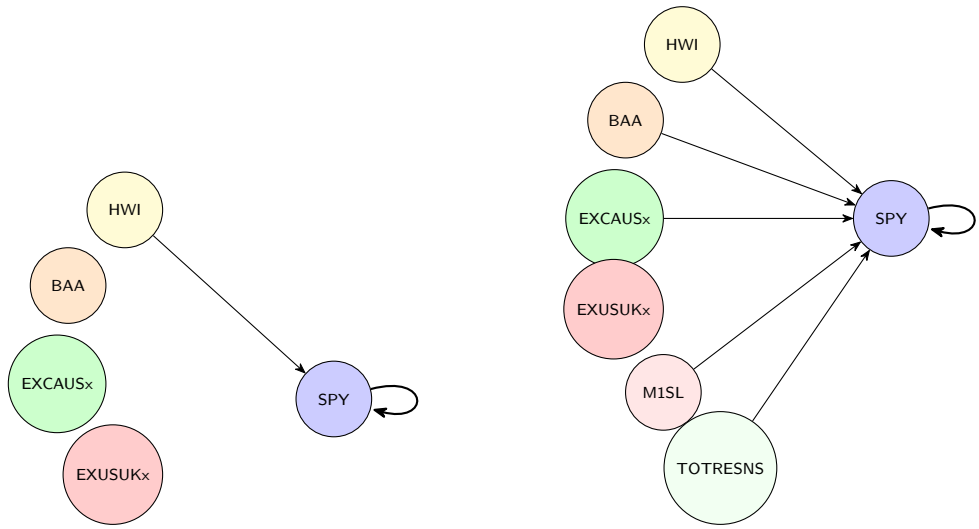


Figure 6: The figure shows the rolling RMSE for SPY return forecasts using four feature selection methods: PCMCI, VAR-LiNGAM, SFSCV B SVM, and SFSCV F Lin (top panel). The middle and bottom panels report the RMSE differences between SFSCV B SVM and the two IBFS benchmarks, VAR-LiNGAM and PCMCI, respectively. Positive values in the difference plots indicate superior performance of the IBFS methods. Shaded regions denote crisis periods.

The directed graphs in Figure 7 summarize the causal relationships uncovered by PCMCI and Multivariate Granger for predicting monthly log returns of the SPY ETF. A key result across both methods is the presence of a self-loop on the SPY node, signaling a strong autoregressive component in SPY returns. This highlights the persistent and predictable structure embedded in equity return dynamics, where past values of the SPY ETF contain information relevant for forecasting its future movements. The autoregressive link is especially important in financial applications, as it provides a baseline signal that can be leveraged for short-horizon timing strategies.

In addition to this autoregressive structure, both methods identify the Help-Wanted Index (HWI) as a causal driver of SPY returns, suggesting that U.S. labor demand conditions influence equity markets in a predictive manner. Multivariate Granger uncovers a broader causal graph, selecting five additional predictors: the BAA corporate bond yield (credit risk), M1SL money supply (liquidity), the trade-weighted exchange rates EXUSUKx and EXCAUSx (external sector), and total reserves held by the Federal Reserve (monetary stance). In contrast, PCMCI favors a more parsimonious representation, identifying only HWI alongside the SPY's own

lag. Together, these results underscore the joint importance of internal market dynamics and selected macro-financial indicators in driving ETF return predictability.



(a) PCMCI (b) Multivariate Granger

Figure 7: The figure presents a summary graph comparing the output of various causal discovery methods applied to the ETF case study at the last time step of the recursive out-of-sample exercise. Each subfigure shows the directed edges inferred by a specific method, where nodes represent candidate predictors and edges indicate discovered causal relationships toward the target variable `SPY` (log return of the SPY ETF). Self-loops represent identified autocorrelation. Top two methods are included.

6.2. Economic value

Currency Case Study: In this section, we assess the economic value of forecasts by evaluating the performance of trading strategies derived from each model's directional signals. Table 5 presents the Sharpe and Sortino ratios for both normal and crisis periods, along with the corresponding generalization gaps computed as the difference in risk-adjusted returns between crisis and normal periods. Smaller (i.e., less negative or positive) gaps signal greater robustness of portfolio performance under heightened volatility, reflecting a model's ability to maintain return quality during regime shifts.

We report both the actual levels of the Sharpe and Sortino ratios and the generalization gaps to provide a complete picture of each method's performance. The absolute levels are included for transparency and to allow practitioners to assess the quality of the trading signals in each regime. However, we emphasize that our objective is not to build an optimal trading algorithm. Had that been the goal, we would have employed more

sophisticated prediction models, position sizing rules, and risk management overlays. Instead, as discussed in Section 3, we deliberately rely on a simple linear forecasting model to isolate the effect of the feature selection step. The trading strategy serves the same purpose: it translates directional forecasts into an economic metric that complements the statistical evaluation, not as an end in itself but as a lens through which to compare IBFS and PBFS methods. In this context, the generalization gaps are the primary basis for comparison, as they directly measure robustness to regime changes, which is the central research question of this paper. A method that maintains or improves its risk adjusted performance during crises, reflected in a gap closer to zero or positive, provides stronger evidence of forecast stability than a method that achieves high absolute returns in normal periods but deteriorates sharply under stress.

Among PBFS methods, most variants suffer considerable deterioration in both Sharpe and Sortino metrics during crises, with typical Sharpe gaps ranging from -0.19 to -0.82 and Sortino gaps from -0.21 to -1.01. The best performing PBFS method is SFSCV F Lin, which achieves a positive Sharpe gap of $+0.05$ and a positive Sortino gap of $+0.05$, indicating slight improvement during crises. However, its absolute performance levels remain modest (Sharpe of -0.16 in normal periods and -0.11 in crises), suggesting that its resilience is related more to low exposure than to predictive skill.

By contrast, IBFS methods exhibit clear advantages across both metrics in levels and in gaps. SeqICP achieves the highest Sharpe gap ($+0.34$) and the highest Sortino gap ($+1.45$), with positive absolute Sharpe ratios in both normal (0.16) and crisis (0.50) periods and Sortino ratios of 0.26 and 1.71, respectively. PCMCI and VAR-LiNGAM also deliver positive Sharpe and Sortino ratios across both regimes, with gaps of $+0.10$ and $+0.08$ for the Sharpe ratio and $+0.16$ and $+0.22$ for the Sortino ratio, respectively. These results demonstrate that causality inspired methods not only preserve predictive robustness but also translate into economically meaningful outcomes under stress. Ultimately, this supports our hypothesis that IBFS methods provide more reliable signals for tactical allocation, especially when return distributions shift abruptly.

Model	Sharpe Ratio		Sortino Ratio		Generalization Gap	
	Normal	Crisis	Normal	Crisis	Sharpe Gap	Sortino Gap
SFSCV B Lin	-0.15	-0.34	-0.16	-0.37	-0.19	-0.21
SFSCV B RF	-0.10	-0.31	-0.10	-0.35	-0.21	-0.25
SFSCV B SVM	-0.40	-0.84	-0.41	-0.99	-0.44	-0.58
SFSCV F Lin	-0.16	-0.11	-0.18	-0.13	**0.05**	**0.05**
SFSCV F RF	-0.64	-1.46	-0.62	-1.63	-0.82	-1.01
SFSCV F SVM	-0.48	-0.88	-0.48	-0.97	-0.40	-0.49
SFSTSCV B Lin	-0.22	-0.25	-0.24	-0.30	-0.03	-0.06
SFSTSCV B RF	-0.38	-0.58	-0.39	-0.66	-0.20	-0.27
SFSTSCV B SVM	-0.39	-0.98	-0.38	-1.09	-0.59	-0.71
SFSTSCV F Lin	-0.73	-1.20	-0.72	-1.35	-0.47	-0.63
SFSTSCV F RF	-0.26	-0.30	-0.27	-0.33	-0.04	-0.06
SFSTSCV F SVM	0.00	0.02	0.01	0.04	0.02	0.03
Dynotears	-0.05	-0.22	-0.07	-0.37	-0.17	-0.30
Pairwise Granger	-0.19	-0.50	-0.26	-0.84	-0.31	-0.58
Multivariate Granger	-0.40	-0.35	-0.52	-0.60	0.05	-0.08
SeqICP	0.16	0.50	0.26	1.71	**0.34**	**1.45**
PCMCI	0.07	0.17	0.09	0.25	0.10	0.16
VAR-LiNGAM	0.14	0.22	0.12	0.34	0.08	0.22

Table 5: This table reports annualized Sharpe and Sortino ratios for the non-crisis ("normal") and crisis periods in the Currency case study. The last two columns show the generalization gap for each metric, calculated as the difference between performance in crisis and normal periods. Higher Sharpe and Sortino ratios indicate better risk-adjusted performance. For the generalization gap, a value closer to zero or positive reflects greater robustness to distributional shifts, indicating that the method maintains or improves its risk-adjusted performance during crisis periods. The best-performing model for each group (PBFS and IBFS) and the overall best models are highlighted in bold.

To evaluate the robustness of risk-adjusted returns across market regimes, we conduct statistical tests comparing the Sharpe ratios of IBFS methods against the PBFS benchmark (SFSCV F RF) during crisis periods in the EUR/CHF forecasting task. We report three complementary tests with progressively stronger corrections for the statistical challenges inherent in comparing Sharpe ratios from dependent, autocorrelated return series.

Table 6 reports results from a standard one-sided Welch's t-test applied to rolling 12 month annualized Sharpe ratios during crisis periods. Under this specification, all six IBFS methods show statistically significant out-performance relative to the PBFS benchmark, with p-values at or below 0.038. SeqICP achieves the largest mean difference (2.369), followed by VAR-LiNGAM (1.959) and Multivariate Granger (1.946). However, rolling Sharpe ratios exhibit substantial autocorrelation by construction, which may inflate test statistics and understate p-values.

To address this concern, Table 7 repeats the Welch's t-test with effective sample sizes corrected for autocorrelation. Under this more conservative specification, only SeqICP ($p = 0.020$) and Multivariate Granger ($p = 0.026$) retain statistical significance at the 5% level. PCMCI falls just below the 10% threshold ($p = 0.093$), while Dynotears ($p = 0.176$), Pairwise

Granger (p = 0.262), and VAR-LiNGAM (p = 0.165) lose significance, reflecting the substantial reduction in effective degrees of freedom induced by the autocorrelation correction. Notably, VAR-LiNGAM's effective sample size drops to approximately 2 observations, producing an extremely wide confidence interval (−4.754 to 8.671) that renders the test uninformative for this method.

Finally, Table 8 reports results from the [83] Sharpe ratio equality test, which operates directly on raw monthly strategy returns rather than rolling Sharpe ratios and obtains standard errors via the delta method with Newey-West HAC covariance estimation. This approach avoids the autocorrelation issues inherent in rolling window statistics while accounting for serial dependence in the return series themselves. Under this specification, all six IBFS methods achieve statistical significance at the 5% level. SeqICP shows the largest SR difference (0.573, p = 0.000), followed by VAR-LiNGAM (0.492, p = 0.017), PCMCI (0.477, p = 0.000), Dynotears (0.362, p = 0.000), Multivariate Granger (0.324, p = 0.005), and Pairwise Granger (0.280, p = 0.005). The uniformly significant results under this robust testing framework provide strong evidence that the superior crisis period Sharpe ratios of IBFS methods are not artifacts of rolling window construction or serial dependence.

Taken together, the three tests paint a consistent picture. SeqICP delivers statistically significant outperformance across all three specifications, providing the strongest evidence that its invariance based feature selection translates into superior risk-adjusted returns during crisis periods. The [83] test, which offers the most robust inference by operating on raw returns with HAC standard errors, confirms significance for all IBFS methods, while the autocorrelation corrected Welch test identifies SeqICP and Multivariate Granger as the most robust performers. As emphasized throughout the paper, our primary focus is on relative performance rather than absolute levels, and the convergence of evidence across multiple testing frameworks reinforces the conclusion that IBFS methods deliver more robust portfolio performance when market conditions are most challenging.

Comparison	Mean Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.860	0.062	1.657	0.017**
Pairwise Granger − SFSCV B SVM	0.777	-0.082	1.636	0.038**
Multivariate Granger − SFSCV B SVM	1.946	1.144	2.748	0.000**
SeqICP − SFSCV B SVM	2.369	1.577	3.161	0.000**
PCMCI − SFSCV B SVM	1.753	0.874	2.631	0.000**
VAR-LiNGAM − SFSCV B SVM	1.959	1.082	2.835	0.000**

Table 6: One-sided Welch's t-test on rolling 12 month annualized Sharpe ratios during crisis periods (EUR/CHF). $H_0\colon \mu^{\text{crisis}}_{\text{IBFS}} = \mu^{\text{crisis}}_{\text{PBFS}}, \quad H_1\colon \mu^{\text{crisis}}_{\text{IBFS}} > \mu^{\text{crisis}}_{\text{PBFS}}$. Positive mean differences indicate IBFS outperformance. 95% confidence intervals. ** denotes significance at the 5% level.

Comparison	Mean Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.860	-1.022	2.741	0.176
Pairwise Granger − SFSCV B SVM	0.777	-1.764	3.318	0.262
Multivariate Granger − SFSCV B SVM	1.946	-0.028	3.920	0.026**
SeqICP − SFSCV B SVM	2.369	0.133	4.605	0.020**
PCMCI − SFSCV B SVM	1.753	-1.004	4.509	0.093
VAR-LiNGAM − SFSCV B SVM	1.959	-4.754	8.671	0.165

Table 7: One-sided Welch's t-test on rolling 12 month annualized Sharpe ratios during crisis periods (EUR/CHF), with effective sample sizes corrected for autocorrelation. $H_0\colon \mu^{\text{crisis}}_{\text{IBFS}} = \mu^{\text{crisis}}_{\text{PBFS}}, \quad H_1\colon \mu^{\text{crisis}}_{\text{IBFS}} > \mu^{\text{crisis}}_{\text{PBFS}}$. Positive mean differences indicate IBFS outperformance. 95% confidence intervals. ** denotes significance at the 5% level.

Comparison	SR Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.362	0.210	0.515	0.000**
Pairwise Granger − SFSCV B SVM	0.280	0.086	0.475	0.005**
Multivariate Granger − SFSCV B SVM	0.324	0.097	0.552	0.005**
SeqICP − SFSCV B SVM	0.573	0.348	0.798	0.000**
PCMCI − SFSCV B SVM	0.477	0.271	0.683	0.000**
VAR-LiNGAM − SFSCV B SVM	0.492	0.087	0.898	0.017**

Table 8: Ledoit and Wolff [83] Sharpe ratio equality test with HAC standard errors applied to monthly strategy returns during crisis periods (EUR/CHF). Unlike the rolling Sharpe ratio tests in Tables 6 and 7, the inputs are raw monthly strategy returns. Sharpe ratios and their difference are computed internally from the return series, with standard errors obtained via the delta method and Newey-West HAC covariance estimation. $H_0\colon \mathrm{SR}_{\text{IBFS}} = \mathrm{SR}_{\text{PBFS}}, \quad H_1\colon \mathrm{SR}_{\text{IBFS}} \neq \mathrm{SR}_{\text{PBFS}}$. Positive SR differences indicate IBFS outperformance. 95% asymptotic confidence intervals. ** denotes significance at the 5% level.

Exchange-traded Funds (ETFs) Case Study: Table 9 presents analogous evidence for the ETF forecasting task. As before, we evaluate IBFS and PBFS methods using Sharpe and Sortino ratios in both normal and crisis periods, as well as their respective generalization gaps. As with the currency case study, the absolute levels are reported for transparency, while the gaps serve as the primary basis for comparing robustness across methods. For the PBFS class, we retain the top-performing method (SFSCV B SVM) as our representative benchmark.

The PBFS benchmark (SFSCV B SVM) achieves a positive Sharpe ratio of 0.34 during normal periods but drops to approximately zero during crises, yielding a Sharpe gap of −0.34 and a Sortino gap of −0.46. This substantial deterioration confirms the pattern observed in the currency case study: purely predictive feature selection methods are vulnerable to distributional shifts.

Among the IBFS models, Pairwise Granger stands out with the best overall performance, achieving a Sharpe gap of 0.00 and a Sortino gap of +0.18, indicating that its risk-adjusted returns are fully preserved and even improve in terms of downside risk during crises. Its absolute levels are also

the highest among all methods (Sharpe of 0.83 in both regimes, Sortino of 1.33 in normal and 1.51 in crisis periods). Multivariate Granger exhibits a similar pattern, with a Sharpe gap of 0.00 and a Sortino gap of +0.15, alongside strong absolute performance (Sharpe of 0.69 in both regimes). VAR-LiNGAM also demonstrates resilience, with a modest Sharpe gap of −0.04 and a positive Sortino gap of +0.08, maintaining Sharpe ratios of 0.69 and 0.65 across normal and crisis periods, respectively. Dynotears shows some deterioration (Sharpe gap of −0.21, Sortino gap of −0.19) but retains positive absolute performance in both regimes. PCMCI suffers a more notable drop, with a Sharpe gap of −0.34 and a Sortino gap of −0.41, representing the weakest IBFS performance in this case study.

These results reinforce our central thesis: under turbulent market conditions, IBFS methods consistently deliver more robust economic outcomes than their PBFS counterparts. In the ETF case study, Pairwise Granger, Multivariate Granger, and VAR-LiNGAM all maintain or improve their risk-adjusted performance during crises, while the PBFS benchmark deteriorates sharply. By emphasizing relative rather than absolute performance, we show that causality informed feature selection offers a meaningful advantage for portfolio decisions in the presence of structural breaks and regime shifts.

Model	Sharpe Ratio		Sortino Ratio		Generalization Gap	
	Normal	Crisis	Normal	Crisis	Sharpe Gap	Sortino Gap
SFSCV B SVM	**0.34**	**-0.00**	**0.46**	**-0.00**	**-0.34**	**-0.46**
Dynotears	0.62	0.41	0.99	0.80	-0.21	-0.19
Pairwise Granger	**0.83**	**0.83**	**1.33**	**1.51**	**0.00**	**0.18**
Multivariate Granger	0.69	0.69	1.09	1.24	**0.00**	0.15
PCMCI	0.48	0.14	0.65	0.24	-0.34	-0.41
VAR-LiNGAM	0.69	0.65	1.00	1.08	-0.04	0.08

Table 9: This table reports annualized Sharpe and Sortino ratios for the non-crisis ("normal") and crisis periods in the ETF case study. The last two columns show the generalization gap for each metric, calculated as the difference between performance in crisis and normal periods. Higher Sharpe and Sortino ratios indicate better risk-adjusted performance. For the generalization gap, a value closer to zero or positive reflects greater robustness to distributional shifts, indicating that the method maintains or improves its risk-adjusted performance during crisis periods. The best-performing model for each group (PBFS and IBFS) and the overall best models are highlighted in bold.

To evaluate the robustness of risk-adjusted returns across market regimes in the ETF case study, we conduct the same three complementary statistical tests as in the currency analysis, comparing IBFS methods against the PBFS benchmark (SFSCV B SVM) during crisis periods.

Table 10 reports results from a standard one-sided Welch's t-test applied to rolling 12 month annualized Sharpe ratios during crisis periods. Under this specification, four of the five IBFS methods show statistically

significant outperformance: Pairwise Granger ($p = 0.000$), Dynotears ($p = 0.001$), Multivariate Granger ($p = 0.001$), and VAR-LiNGAM ($p = 0.005$). PCMCI shows a positive mean difference (0.172) but does not reach significance ($p = 0.229$).

Table 11 repeats the test with effective sample sizes corrected for autocorrelation. Under this more conservative specification, none of the IBFS methods retain statistical significance, with p-values ranging from 0.139 (Pairwise Granger) to 0.417 (PCMCI). The substantial reduction in effective sample sizes, driven by the strong autocorrelation inherent in rolling window statistics, severely limits statistical power and widens confidence intervals.

Table 12 reports results from the [83] test applied to raw monthly strategy returns with HAC standard errors. This approach avoids the autocorrelation issues of rolling window statistics by operating directly on the return series. Consistent with the autocorrelation corrected Welch test, none of the IBFS methods achieve statistical significance at the 5% level. However, all methods except PCMCI show positive SR differences (0.095 to 0.169), indicating economically meaningful outperformance during crises that lacks the statistical power to reject the null.

The ETF results present a weaker statistical case than the currency case study, where SeqICP achieved significance across all three testing frameworks. Two factors contribute to this difference. First, the ETF crisis subsample is shorter, reducing statistical power across all specifications. Second, the PBFS benchmark (SFSCV B SVM) performs poorly in absolute terms, as shown in Table 4, generating strategy returns with high variance that further inflate standard errors. Nevertheless, the direction of the evidence is consistent across all three tests: the majority of IBFS methods show positive Sharpe ratio differences during crises, and the standard Welch test confirms significance for four of five methods before autocorrelation correction. Combined with the prediction metric evidence in Table 4 and the rolling RMSE analysis in Figure 6, the overall picture supports the conclusion that IBFS methods deliver more robust and stable performance under distributional stress, even in the more challenging ETF forecasting environment.

Comparison	Mean Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.767	0.296	1.237	0.001**
Pairwise Granger − SFSCV B SVM	1.092	0.764	1.420	0.000**
Multivariate Granger − SFSCV B SVM	0.715	0.285	1.145	0.001**
PCMCI − SFSCV B SVM	0.172	-0.284	0.627	0.229
VAR-LiNGAM − SFSCV B SVM	0.553	0.137	0.969	0.005**

Table 10: One-sided Welch's t-test on rolling 12 month annualized Sharpe ratios during crisis periods (ETF). $H_0\colon \mu^{\text{crisis}}_{\text{IBFS}} = \mu^{\text{crisis}}_{\text{PBFS}}$, $H_1\colon \mu^{\text{crisis}}_{\text{IBFS}} > \mu^{\text{crisis}}_{\text{PBFS}}$. Positive mean differences indicate IBFS outperformance. 95% confidence intervals. ** denotes significance at the 5% level.

Comparison	Mean Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.767	-1.958	3.491	0.236
Pairwise Granger − SFSCV B SVM	1.092	-1.883	4.067	0.139
Multivariate Granger − SFSCV B SVM	0.715	-0.981	2.412	0.181
PCMCI − SFSCV B SVM	0.172	-1.690	2.033	0.417
VAR-LiNGAM − SFSCV B SVM	0.553	-1.032	2.137	0.221

Table 11: One-sided Welch's t-test on rolling 12 month annualized Sharpe ratios during crisis periods (ETF), with effective sample sizes corrected for autocorrelation. $H_0\colon \mu^{\text{crisis}}_{\text{IBFS}} = \mu^{\text{crisis}}_{\text{PBFS}}$, $H_1\colon \mu^{\text{crisis}}_{\text{IBFS}} > \mu^{\text{crisis}}_{\text{PBFS}}$. Positive mean differences indicate IBFS outperformance. 95% confidence intervals. ** denotes significance at the 5% level.

Comparison	SR Difference	CI Lower	CI Upper	P-Value
Dynotears − SFSCV B SVM	0.095	-0.178	0.368	0.495
Pairwise Granger − SFSCV B SVM	0.169	-0.140	0.478	0.283
Multivariate Granger − SFSCV B SVM	0.164	-0.131	0.459	0.276
PCMCI − SFSCV B SVM	-0.000	-0.263	0.263	0.998
VAR-LiNGAM − SFSCV B SVM	0.128	-0.150	0.406	0.367

Table 12: Ledoit and Wolf [83] Sharpe ratio equality test with HAC standard errors applied to monthly strategy returns during crisis periods (ETF). The inputs are raw monthly strategy returns. Sharpe ratios and their difference are computed internally from the return series, with standard errors obtained via the delta method and Newey-West HAC covariance estimation. $H_0\colon \mathrm{SR}_{\text{IBFS}} = \mathrm{SR}_{\text{PBFS}}$, $H_1\colon \mathrm{SR}_{\text{IBFS}} \neq \mathrm{SR}_{\text{PBFS}}$. Positive SR differences indicate IBFS outperformance. 95% asymptotic confidence intervals. ** denotes significance at the 5% level.

6.3. Extended Benchmarks

6.4. Note on the Role of Sampling Frequency and Model Complexity

A potential concern with our empirical design is that the monthly sampling frequency limits the number of effective observations available for estimation, which may place nonparametric and machine learning methods at a structural disadvantage relative to parametric models. If this were the case, part of the observed performance gap between IBFS and PBFS

methods could reflect differences in statistical efficiency rather than the intrinsic value of causality inspired feature selection.

We acknowledge this concern but argue that it does not undermine our conclusions for several reasons. First, a growing body of evidence demonstrates that machine learning methods can and do outperform linear models on monthly financial and macroeconomic data. [34] show that tree based methods and neural networks achieve substantial improvements over linear benchmarks in predicting monthly equity returns using a large cross section of firm characteristics. [84] find that machine learning methods, including random forests and elastic net, deliver superior inflation forecasts relative to traditional econometric models using monthly macroeconomic data. [85] conduct a systematic evaluation of machine learning methods for macroeconomic forecasting using the FRED-MD dataset at monthly frequency and find that nonlinearities significantly improve forecasting accuracy for several key variables, with improvements reaching 23% for longer horizons. [86] similarly demonstrate that penalized regression methods outperform standard approaches on monthly FRED-MD data. These findings indicate that the monthly frequency per se does not preclude machine learning methods from outperforming linear alternatives, provided sufficient predictor richness and appropriate regularization are employed.

Second, and more importantly, the PBFS class in our study is not composed exclusively of nonparametric or machine learning models. The sequential feature selection (SFS) procedure is paired with three different downstream estimators: a linear model (Lin), a random forest (RF), and a support vector machine (SVM). The linear variant (SFSCV F Lin, SFSTSCV B Lin) operates within the same parametric framework as the OLS model used in the IBFS pipeline, ensuring a fair comparison that is not confounded by differences in model complexity. Moreover, SVM with a linear kernel occupies an intermediate position between fully parametric and nonparametric methods, offering regularization without the high dimensional flexibility of ensemble methods. The fact that the performance gap between IBFS and PBFS persists across all three model classes, including the linear variant, suggests that the observed advantage is attributable to the feature selection criterion (invariance versus prediction) rather than to differences in estimator flexibility.

Third, the IBFS methods themselves operate under the same sample size constraints as the PBFS alternatives. Both classes of methods receive the same monthly observations and must estimate their respective selection criteria from the same limited data. If small samples were the primary driver of PBFS underperformance, one would expect IBFS methods to suffer similarly, particularly those that require estimating complex graphical structures (e.g., Dynotears, PCMCI). The fact that IBFS methods achieve superior robustness despite facing identical data limitations reinforces the

interpretation that the advantage stems from the invariance principle itself, not from differential sensitivity to sample size.

7. Conclusion

In this study, we systematically evaluated the impact of causality-inspired, invariance-based feature selection methods versus traditional predictive-based approaches for forecasting economic and financial time series. Our empirical results, spanning currency and ETF markets and covering both normal and crisis regimes, show that invariance-based causal methods consistently deliver more robust, stable, and economically valuable forecasts, especially during periods of heightened market volatility. While some predictive-based feature selection can adjust to changing regimes, they often do so from a weaker predictive base and exhibit greater vulnerability to structural breaks. By focusing on features and relationships that are stable across environments, causal methods provide resilience not only in terms of statistical performance but also in risk-adjusted economic outcomes. These findings underscore the practical utility of integrating causal reasoning into financial modeling, particularly for investors and practitioners who seek reliable signals during uncertain conditions and structural market shifts. Future work may extend this framework to other asset classes and examine more sophisticated forms of causal discovery.

During the preparation of this work, the author(s) used Chatgpt in order to polish the writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

[1] M. K. Brunnermeier, S. Nagel, L. H. Pedersen, Carry Trades and Currency Crashes, NBER Working Paper 14473, National Bureau of Economic Research, 2008. URL: https://www.nber.org/papers/w14473. doi:10.3386/w14473, revised December 2013.

[2] T. J. Moskowitz, Y. H. Ooi, L. H. Pedersen, Time series momentum, Journal of Financial Economics 104 (2012) 228–250.

[3] R. Molenaar, et al., Empirical evidence on the stock–bond correlation, The Journal of Portfolio Management 50 (2024) 45–63.

[4] G. I. Webb, L. K. Lee, B. Goethals, F. Petitjean, Analyzing concept drift and shift from sample data, Data Mining and Knowledge Discovery 32 (2018) 1179–1199. URL: https://doi.org/10.1007/s10618-018-0554-1. doi:10.1007/s10618-018-0554-1.

[5] S. Makridakis, E. Spiliotis, V. Assimakopoulos, The m4 competition: Results, findings, conclusion and way forward, International Journal of Forecasting (2018).

[6] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, Journal of Machine Learning Research 3 (2003) 1157–1182.

[7] C. Molnar, Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, Self-published, 2022. https://christophm.github.io/interpretable-ml-book/.

[8] Y. Bengio, A. Courville, P. Vincent, Representation learning: A review and new perspectives, IEEE Transactions on Pattern Analysis and Machine Intelligence 35 (2013) 1798–1828.

[9] C. Rudin, Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead, Nature Machine Intelligence 1 (2019) 206–215.

[10] J. H. Stock, M. W. Watson, Macroeconomic forecasting using diffusion indexes, Journal of Business & Economic Statistics 20 (2002) 147–162.

[11] S. Ng, Variable selection in predictive regressions, in: Handbook of Economic Forecasting, volume 2, Elsevier, 2013, pp. 752–789.

[12] R. Tibshirani, Regression shrinkage and selection via the lasso, Journal of the Royal Statistical Society: Series B (Methodological) 58 (1996) 267–288.

[13] A. Belloni, V. Chernozhukov, ℓ_1-penalized quantile regression in high-dimensional sparse models, The Annals of Statistics 39 (2011) 82–130.

[14] J. Peters, D. Janzing, B. Schölkopf, Elements of Causal Inference: Foundations and Learning Algorithms, MIT Press, 2017.

[15] D. Janzig, Causal regularization, in: Advances in Neural Information Processing Systems, volume 32, 2019. URL: https://proceedings.neurips.cc/paper_files/paper/2019/file/2172fde49301047270b2897085e4319d-Paper.pdf.

[16] T. Kyono, Y. Zhang, M. van der Schaar, Castle: Regularization via auxiliary causal graph discovery, in: Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS'20), Curran Associates Inc., Red Hook, NY, USA, 2020, pp. 1–12.

[17] D. Lopez-Paz, R. Nishihara, S. Chintala, B. Schölkopf, L. Bottou, Discovering causal signals in images, in: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 58–66. URL: https://api.semanticscholar.org/CorpusID:1847130.

[18] N. Pfister, P. Bühlmann, J. Peters, Invariant causal prediction for sequential data, Journal of the American Statistical Association 114 (2019) 1264–1276. URL: https://doi.org/10.1080/01621459.2018.1491403. doi:10.1080/01621459.2018.1491403.

[19] R. Godahewa, C. Bergmeir, Z. Erkin Baz, C. Zhu, Z. Song, S. García, D. Benavides, On forecast stability, International Journal of Forecasting 41 (2025) 1539–1558. URL: https://www.sciencedirect.com/science/article/pii/S0169207025000068. doi:https://doi.org/10.1016/j.ijforecast.2025.01.006.

[20] L. W. Ancel, W. Fontana, Plasticity, evolvability, and modularity in rna, Journal of Experimental Zoology 288 (2000) 242–283. doi:10.1002/1097-010x(20001015)288:3<242::aid-jez5>3.0.co;2-o.

[21] A. Wagner, Robustness and evolvability: a paradox resolved, Proceedings of the Royal Society B: Biological Sciences 275 (2008) 91–100. doi:10.1098/rspb.2007.1137.

[22] P. Bühlmann, Invariance, causality and robustness, Statistical Science 35 (2020) 404–426. doi:10.1214/19-STS721.

[23] J. Peters, P. Bühlmann, N. Meinshausen, Causal inference by using invariant prediction: identification and confidence intervals, Journal of the Royal Statistical Society: Series B (Statistical Methodology) 78 (2016) 947–1012.

[24] M. W. McCracken, S. Ng, Fred-md: A monthly database for macroeconomic research, Journal of Business & Economic Statistics 34 (2016) 574–589. Special Issue on Big Data.

[25] R. S. Tsay, Analysis of Financial Time Series, 2nd ed., Wiley-Interscience, 2005.

[26] R. Cont, Empirical properties of asset returns: stylized facts and statistical issues, Quantitative Finance 1 (2001) 223.

[27] P. Whittle, Hypothesis testing in time series analysis, 1951.

[28] A. A. Ariyo, A. O. Adewumi, C. K. Ayo, Stock price prediction using the arima model, in: 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation, IEEE, 2014, pp. 106–112.

[29] C. A. Sims, Macroeconomics and reality, Econometrica: Journal of the Econometric Society (1980) 1–48.

[30] J. H. Stock, M. W. Watson, Vector autoregressions, Journal of Economic Perspectives 15 (2001) 101–115.

[31] B. M. Henrique, V. A. Sobreiro, H. Kimura, Literature review: Machine learning techniques applied to financial market prediction, Expert Systems with Applications 124 (2019) 226–251.

[32] M. M. Kumbure, C. Lohrmann, P. Luukka, J. Porras, Machine learning techniques and data for stock market forecasting: A literature review, Expert Systems with Applications 197 (2022) 116659.

[33] C. Krauss, X. A. Do, N. Huck, Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the s&p 500, European Journal of Operational Research 259 (2017) 689–702.

[34] S. Gu, B. Kelly, D. Xiu, Empirical asset pricing via machine learning, The Review of Financial Studies 33 (2020) 2223–2273. doi:10.1093/rfs/hhaa009.

[35] J. Y. Campbell, R. J. Shiller, The dividend-price ratio and expectations of future dividends and discount factors, The Review of Financial Studies 1 (1988) 195–228.

[36] M. S. Rozeff, Dividend yields are equity risk premiums, Journal of Portfolio Management (1984) 68–75.

[37] S. P. Kothari, J. Shanken, Book-to-market, dividend yield, and expected market returns: A time-series analysis, Journal of Financial Economics 44 (1997) 169–203.

[38] A. Ang, G. Bekaert, Stock return predictability: Is it there?, The Review of Financial Studies 20 (2007) 651–707.

[39] J. Y. Campbell, M. Yogo, Efficient tests of stock return predictability, Journal of Financial Economics 81 (2006) 27–60.

[40] C. S. Asness, R. B. Porter, R. L. Stevens, Predicting stock returns using industry-relative firm characteristics, Available at SSRN 213872 (2000).

[41] A. W. Lo, A. C. MacKinlay, Stock market prices do not follow random walks: Evidence from a simple specification test, The Review of Financial Studies 1 (1988) 41–66.

[42] A. W. Lo, H. Mamaysky, J. Wang, Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation, The Journal of Finance 55 (????).

[43] S. N. Neftci, Naive trading rules in financial markets and wiener-kolmogorov prediction theory: A study of "technical analysis", Journal of Business (1991) 549–571.

[44] C. J. Neely, D. E. Rapach, J. Tu, G. Zhou, Forecasting the equity risk premium: the role of technical indicators, Management Science 60 (2014) 1772–1791.

[45] Q. Lin, Technical analysis and stock return predictability: An aligned approach, Journal of Financial Markets 38 (2018) 103–123.

[46] C. Çakmaklı, D. van Dijk, Getting the most out of macroeconomic information for predicting excess stock returns, International Journal of Forecasting 32 (2016) 650–668.

[47] E. F. Fama, G. W. Schwert, Asset returns and inflation, Journal of Financial Economics 5 (1977) 115–146.

[48] L. Li, Macroeconomic factors and the correlation of stock and bond returns, Available at SSRN 363641 (2002).

[49] M. J. Flannery, A. A. Protopapadakis, Macroeconomic factors do influence aggregate stock returns, The Review of Financial Studies 15 (2002) 751–782.

[50] T. Chordia, A. Subrahmanyam, Order imbalance and individual stock returns: Theory and evidence, Journal of Financial Economics 72 (2004) 485–518.

[51] T. Chordia, R. Roll, A. Subrahmanyam, Order imbalance, liquidity, and market returns, Journal of Financial Economics 65 (2002) 111–130.

[52] Y. Lu, G. Reinert, M. Cucuringu, Trade co-occurrence, trade flow decomposition and conditional order imbalance in equity markets, Quantitative Finance (2024) 1–31.

[53] Y. Lu, G. Reinert, M. Cucuringu, Co-trading networks for modeling dynamic interdependency structures and estimating high-dimensional covariances in us equity markets, arXiv preprint arXiv:2302.09382 (2023).

[54] R. Cont, M. Cucuringu, C. Zhang, Cross-impact of order flow imbalance in equity markets, Quantitative Finance 23 (2023) 1373–1393. doi:10.1080/14697688.2023.2236159.

[55] I. Welch, A. Goyal, A comprehensive look at the empirical performance of equity premium prediction, The Review of Financial Studies 21 (2008) 1455–1508.

[56] J. Y. Campbell, S. B. Thompson, Predicting excess stock returns out of sample: Can anything beat the historical average?, The Review of Financial Studies 21 (2008) 1509–1531.

[57] C.-F. Tsai, Y.-C. Hsiao, Combining multiple feature selection methods for stock prediction: Union, intersection, and multi-intersection approaches, Decision Support Systems 50 (2010) 258–269.

[58] S. Thawornwong, D. Enke, The adaptive selection of financial and economic variables for use with artificial neural networks, Neurocomputing 56 (2004) 205–232.

[59] Y. Zhang, F. Ma, Y. Wang, Forecasting crude oil prices with a large set of predictors: Can lasso select powerful predictors?, Journal of Empirical Finance 54 (2019) 97–117.

[60] A. Chinco, A. D. Clark-Joseph, M. Ye, Sparse signals in the cross-section of returns, The Journal of Finance 74 (2019) 449–492.

[61] M. A. Hall, Correlation-based feature selection for machine learning, Ph.D. thesis, The University of Waikato, 1999.

[62] J. Pearl, Causality, Cambridge University Press, 2009.

[63] N. Pfister, E. G. Williams, J. Peters, R. Aebersold, P. Bühlmann, Stabilizing variable selection and regression, The Annals of Applied Statistics 15 (2021) 1220–1246. URL: https://doi.org/10.1214/21-AOAS1487. doi:10.1214/21-AOAS1487.

[64] T. Haavelmo, The probability approach in econometrics, Econometrica: Journal of the Econometric Society (1944) iii–115.

[65] P. Spirtes, C. N. Glymour, R. Scheines, Causation, prediction, and search, MIT Press, 2000.

[66] L. Kilian, H. Lütkepohl, Structural Vector Autoregressive Analysis, Cambridge University Press, Cambridge, 2017.

[67] J. Runge, P. Nowack, M. Kretschmer, S. Flaxman, D. Sejdinovic, Detecting and quantifying causal associations in large nonlinear time series datasets, Science Advances 5 (2019) eaau4996.

[68] A. Hyvärinen, K. Zhang, S. Shimizu, P. O. Hoyer, Estimation of a structural vector autoregression model using non-gaussianity, Journal of Machine Learning Research 11 (2010).

[69] C. W. Granger, Investigating causal relations by econometric models and cross-spectral methods, Econometrica: Journal of the Econometric Society 37 (1969) 424–438.

[70] R. Pamfil, N. Sriwattanaworachai, S. Desai, P. Pilgerstorfer, P. Beaumont, K. Georgatzis, B. Aragam, Dynotears: Structure learning from time-series data, in: Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics (AISTATS), volume 108 of *Proceedings of Machine Learning Research*, PMLR, 2020, pp. 1595–1605.

[71] H. Lütkepohl, New Introduction to Multiple Time Series Analysis, Springer, 2005. URL: https://EconPapers.repec.org/RePEc:spr:sprbok:978-3-540-27752-1.

[72] P. Spirtes, C. Glymour, An algorithm for fast recovery of sparse causal graphs, Social Science Computer Review 9 (1991) 62–72.

[73] I. Guyon, A. Elisseeff, An introduction to variable and feature selection, Journal of Machine Learning Research 3 (2003) 1157–1182.

[74] A. W. Whitney, A direct method of nonparametric measurement selection, IEEE Transactions on Computers 100 (1971) 1100–1103.

[75] S. M. Goldfeld, R. E. Quandt, A markov model for switching regressions, Journal of Econometrics 1 (1973) 3–15.

[76] A. C. Harvey, Dynamic Models for Volatility and Heavy Tails: With Applications to Financial and Economic Time Series, Cambridge University Press, 2013.

[77] L. Breiman, Random forests, Machine Learning 45 (2001) 5–32.

[78] J. H. Friedman, Greedy function approximation: A gradient boosting machine, Annals of Statistics 29 (2001) 1189–1232.

[79] R. P. Masini, M. C. Medeiros, E. F. Mendes, Machine learning advances for time series forecasting, Journal of Economic Surveys 37 (2023) 76–111. doi:10.1111/joes.12429.

[80] S. C. Ludvigson, S. Ng, Macro factors in bond risk premia, The Review of Financial Studies 22 (2009) 5027–5067. doi:10.1093/rfs/hhp081.

[81] S. Smeekes, E. Wijler, Macroeconomic forecasting using penalized regression methods, International Journal of Forecasting 34 (2018) 408–430. URL: https://www.sciencedirect.com/science/article/pii/S0169207018300074. doi:https://doi.org/10.1016/j.ijforecast.2018.01.001.

[82] J. Bae, Factor-augmented forecasting in big data, International Journal of Forecasting 40 (2024) 1660–1688. URL: https://www.sciencedirect.com/science/article/pii/S0169207024000098. doi:https://doi.org/10.1016/j.ijforecast.2024.02.004.

[83] O. Ledoit, M. Wolf, Robust performance hypothesis testing with the Sharpe ratio, Journal of Empirical Finance 15 (2008) 850–859. doi:10.1016/j.jempfin.2008.03.002.

[84] M. C. Medeiros, G. F. R. Vasconcelos, Á. Veiga, E. Zilberman, Forecasting inflation in a data-rich environment: The benefits of machine learning methods, Journal of Business & Economic Statistics 39 (2021) 98–119. doi:10.1080/07350015.2019.1637745.

[85] P. G. Coulombe, M. Leroux, D. Stevanovic, S. Surprenant, How is machine learning useful for macroeconomic forecasting?, Journal of Applied Econometrics 37 (2022) 920–964. doi:10.1002/jae.2910.

[86] S. Smeekes, E. Wijler, Macroeconomic forecasting using penalized regression methods, International Journal of Forecasting 34 (2018) 408–430. doi:10.1016/j.ijforecast.2018.01.001.

Appendix A. Hyperparameter Selection and Walk-Forward Evaluation

This appendix details the hyperparameter spaces, tuning procedures, and walk-forward evaluation protocol employed across all forecasting models considered in this paper.

Appendix A.1. Walk-Forward Evaluation Protocol

All models are evaluated using an expanding-window (walk-forward) scheme with a fixed start date. Let T_0 denote the length of the initial estimation window. At each forecasting step $s = 0, 1, \ldots,$ the training sample spans observations $\{1, \ldots, T_0 + s\}$, and a single one-step-ahead forecast is produced for period $T_0 + s + 1$. The estimation window expands as s increases, while the origin remains fixed. This design reflects a realistic investor setting in which all available history is used for model calibration at each point in time.

Within each training window, hyperparameters are selected via *time-series cross-validation* (TSCV) using $K = 5$ sequential folds generated by `TimeSeriesSplit` from `scikit-learn`. This ensures that validation data always follows training data chronologically, preserving the temporal ordering of observations and avoiding look-ahead bias.

For models requiring hyperparameter search, we adopt a *random search* strategy in which $N_{\text{cand}} = 50$ configurations are sampled uniformly from the candidate grid and evaluated via TSCV. The configuration minimizing the average mean squared error across folds is selected. The final model is then re-estimated on the full training window using the selected hyperparameters, and the one-step-ahead forecast is recorded.

All target and predictor variables are standardized to zero mean and unit variance using moments computed exclusively from the training window at each step. Test observations are transformed using the same training-window moments.

Appendix A.2. Linear Penalized Regression Models

Appendix A.2.1. Ridge Regression

The Ridge estimator minimizes the penalized objective

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right\},$$

where $\lambda > 0$ is the regularization parameter. We consider two strategies for constructing the candidate grid over λ:

Data-driven grid (`glmnet`).. Following the convention of **?**], we compute

$$\lambda_{\max} = \frac{1}{n} \max_j |\mathbf{x}_j^\top \mathbf{y}|, \qquad \lambda_{\min} = 10^{-4} \cdot \lambda_{\max},$$

and construct a grid of 200 values log-spaced between $\lambda_{\max}$ and $\lambda_{\min}$. Since `scikit-learn`'s Ridge implementation minimizes $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha\|\boldsymbol{\beta}\|_2^2$, we set $\alpha = n\lambda$.

Classical grid.. A fixed grid of 200 values log-spaced between 10^{-4} and 10^4 is used, independent of the data.

From either grid, $N_{\text{cand}} = 50$ candidates are drawn uniformly at random and evaluated via TSCV with $K = 5$ folds.

Appendix A.2.2. Lasso Regression

The Lasso estimator minimizes

$$\hat{\boldsymbol{\beta}}_{\text{lasso}} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}.$$

The data-driven grid follows the same `glmnet` construction as Ridge, with $\lambda_{\max}$ representing the smallest penalty that sets all coefficients to zero. Since `scikit-learn`'s Lasso implementation uses the same $1/(2n)$ normalization as `glmnet`, the regularization parameter maps directly as $\alpha = \lambda$. The classical grid spans $[10^{-6}, 10^2]$ to reflect the typically smaller scale of ℓ_1 penalties. The coordinate descent solver is run with a maximum of 50,000 iterations and a convergence tolerance of 10^{-4}.

Appendix A.3. Tree-Based Ensemble Models

Appendix A.3.1. Random Forest

The Random Forest hyperparameter space follows the conventions established in the machine learning for asset pricing literature, particularly **?**], **?**], and **?**]. The candidate grid is specified in Table A.13.

Table A.13: Random Forest hyperparameter space

Hyperparameter	Candidate values
`n_estimators`	$\{100, 300, 500, 1000\}$
`max_depth`	$\{1, 2, 3, 4, 5, 6, \text{None}\}$
`max_features`	$\{1/3, 1/2, \sqrt{p}, \log_2 p, 1.0\}$
`min_samples_split`	$\{2, 5, 10, 20\}$
`min_samples_leaf`	$\{1, 2, 5, 10\}$
`bootstrap`	`True`

The emphasis on shallow trees (small `max_depth`) reflects the well-documented finding that financial return prediction operates in a low signal-to-noise environment, where deep trees tend to overfit [?]. The `max_features` parameter controls the degree of feature randomization at each split, with $p/3$ and $\sqrt{p}$ being the most common academic defaults.

Appendix A.3.2. Gradient Boosted Trees

The Gradient Boosted Trees (GBT) hyperparameter space draws on ?], ?], ?], and Medeiros et al. [84]. The candidate grid is specified in Table A.14.

Table A.14: Gradient Boosted Trees hyperparameter space

Hyperparameter	Candidate values
`n_estimators`	$\{100, 300, 500, 1000\}$
`max_depth`	$\{1, 2, 3, 4\}$
`learning_rate`	$\{0.001, 0.005, 0.01, 0.05, 0.1\}$
`subsample`	$\{0.5, 0.7, 0.8, 1.0\}$
`min_samples_split`	$\{2, 5, 10, 20\}$
`min_samples_leaf`	$\{1, 2, 5, 10\}$
`max_features`	$\{\sqrt{p}, \log_2 p, 0.5, 1.0\}$
`loss`	Least squares

The key design principle is to combine a large number of weak learners (`n_estimators`) with strong shrinkage (low `learning_rate`) and shallow interaction depth (`max_depth` ≤ 4). Stochastic gradient boosting (`subsample` < 1) reduces variance and has been shown to improve out-of-sample performance in macroeconomic and financial forecasting applications [?].

Appendix A.4. Markov-Switching Dynamic Regression

The Markov-Switching model is estimated via maximum likelihood using the EM algorithm as implemented in `statsmodels`. The number of regimes K is selected from the candidate set $\{2, 3\}$ by minimizing the Bayesian Information Criterion (BIC) at each estimation window. The model specification includes regime-specific intercepts (`trend='c'`), regime-specific regression coefficients (`switching_exog=True`), and regime-specific error variances (`switching_variance=True`). The EM algorithm is run for a maximum of 200 iterations. Regime probabilities used for forecasting are obtained from the filtered (causal) marginal probabilities to avoid look-ahead bias.

Appendix A.5. Score-Driven (GAS) Models

The Generalized Autoregressive Score models are estimated via maximum likelihood using basin-hopping global optimization as implemented in the `PyTimeVar` package [**?**]. Two distributional specifications are considered: Gaussian ($\mathcal{N}$-GAS) and Student-t (t-GAS). The feature vector is restricted to an intercept and a single lag of the target variable, reflecting the practical constraints of score-driven estimation in finite samples. The static parameters $(\boldsymbol{\omega}, \mathbf{A}, \mathbf{B})$ governing the parameter dynamics are estimated from the data, with diagonal $\mathbf{A}$ and $\mathbf{B}$ matrices. Input features are clipped to $[-5, 5]$ in standardized units to prevent numerical overflow during likelihood evaluation. No additional hyperparameter tuning is performed beyond the distributional choice.

Appendix A.6. Summary

Table A.15 provides an overview of the tuning methodology across all models.

Table A.15: Summary of hyperparameter tuning across models

Model	Selection method	Candidates	Criterion
Ridge	Random search + TSCV	50	MSE
Lasso	Random search + TSCV	50	MSE
Random Forest	Random search + TSCV	50	MSE
Gradient Boosted Trees	Random search + TSCV	50	MSE
Markov-Switching	Grid search	$\{2, 3\}$ regimes	BIC
$\mathcal{N}$-GAS / t-GAS	MLE (basin-hopping)	—	Log-likelihood